# A ZOO Deep Learning Survey

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## Abstract

Zero-order optimization (ZOO) trains a network from loss evaluations alone, without back-propagating errors: the natural training rule for neuromorphic and analog hardware, non-differentiable systems, biologically plausible learning and memory-constrained fine-tuning. Many ZOO rules exist but are rarely compared under one budget on the same networks. We benchmark nine backprop-free training rules—SPSA and its curvature-preconditioned variant 1.5-SPSA, random search, classic, activity-based, iterative and decorrelated node perturbation, curvature-preconditioned activity perturbation (1.5-ANP), and the forward-forward algorithm—against backpropagation on four ≈40k-parameter architectures (an MLP on MNIST, a ConvNet on CIFAR-10, a character-level LSTM and Transformer on Penn Treebank), with the same forward passes per step, one GPU-hour, the same Adam update and a probe scale tied to the learning rate. A screening sweep over five scales selects each method's scale per task; the selected configurations are replicated with three seeds, evaluated every minute and reported at their best validation checkpoint and at the end of the hour, with a backpropagation reference at the same granularity, a grid that decouples the probe scale from the step size, the two curvature-preconditioned rules under their own recipes, and an estimator-alignment measurement. The screening's first lesson is that the step size, not the probe scale, decides: at $\varepsilon = 10^{-4}$ every probe rule improves and the weight-space rules close most of their gap on the image tasks. Iterative node perturbation (INP) is the most consistent zero-order rule: best mean rank, best probe method on both character-level models (2.82 and 3.10 bits per character against 1.78 and 1.82 for backpropagation) and tied for best on the image tasks with classic and activity-based node perturbation (96.9 % on MNIST and 63 % on CIFAR-10, against 97.1 and 69.2 % for backpropagation). Curvature preconditioning leaves SPSA unchanged under Adam and under SGD; 1.5-ANP is usable on the sequence models where ANP is not, but its $\alpha = 0$ control shows that the credit is due to its normalisation and token-local attribution, not to the curvature weighting. Code and run logs: https://github.com/Fchaubard/zero-order-ablations.

## 1 Introduction

Backpropagation [Rumelhart et al., 1986] is the default way to train neural networks, but it is not always available. Neuromorphic and analog accelerators implement a forward pass natively and may have no mechanism to transport exact error signals backwards through the substrate; systems with non-differentiable components (simulators, hardware in the loop, discrete decisions, black-box APIs) expose only a loss value; neuroscience asks which learning rules a brain could plausibly implement with local information; and fine-tuning of very large models is increasingly limited by the activation memory that backpropagation must store [Malladi et al., 2023]. In all of these settings the learner can evaluate the loss at perturbed parameters or activities but cannot differentiate it; *zero-order optimization* (ZOO) estimates a descent direction from such evaluations alone.

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The ZOO literature is old and broad: random search [Rastrigin, 1963, Matyas, 1965, Schumer and Steiglitz, 1968], simultaneous perturbation stochastic approximation [Spall, 1992, 1998], Gaussian smoothing and its convergence theory [Nesterov and Spokoiny, 2017, Duchi et al., 2015, Flaxman et al., 2005], evolution strategies [Wierstra et al., 2014, Hansen et al., 2003, Salimans et al., 2017, Mania et al., 2018], node perturbation [Widrow and Lehr, 1990, Werfel et al., 2003, Fiete and Seung, 2006, Hiratani et al., 2022] and its recent activity-based and decorrelated variants [Dalm et al., 2026], curvature-preconditioned variants [Chaubard, 2025b,a], and local learning rules such as forward-forward [Hinton, 2022]. Each paper demonstrates its method on its own architecture, budget and optimizer; a practitioner who must train without backpropagation is left with the question this paper answers empirically: *which ZOO method wins, on standard small architectures, when every method is given the same budget?*

The protocol (Sections 3 and 4) fixes four ≈40k-parameter models on four standard tasks, gives every probe method the same forward passes per step, one hour of training on one GPU and the same Adam update, ties the probe scale to the learning rate so that a single scalar $\varepsilon$ is swept per method, selects $\varepsilon$ in a single-seed *screening* sweep and *replicates* the selected configurations with three seeds, reported at the best validation checkpoint (the quantity a practitioner would keep) and at the end of the hour. Ablations test the protocol itself: a backpropagation reference at the same granularity, a grid that decouples the probe scale from the step size, the curvature-preconditioned rules under their own recipes, and every estimator's cosine with the true gradient.

**Contributions.** (i) A fully specified two-stage protocol under which ten training rules—backpropagation, SPSA, 1.5-SPSA, random search, four node-perturbation variants, 1.5-ANP and forward-forward—are compared on four architectures with identical forward-pass, wall-clock and optimizer budgets, three seeds, and selection on validation data only. (ii) One implementation of every rule on a shared *perturbable layer* abstraction (Appendix G), so that every method drives the same models, including an LSTM and a pre-LN Transformer; the two curvature-preconditioned rules are implemented from their published code and specification. (iii) The resulting comparison, backpropagation reference, probe-scale / step-size decoupling and estimator-alignment diagnostic (Tables 1, 11, 6 and 7; Figures 1, 3 and 2), with code, configurations and every run log released.

## 2 Related Work

**Weight-space perturbation.** SPSA [Spall, 1992, 1998] estimates a gradient from two loss evaluations at $\boldsymbol{\theta} \pm \sigma \mathbf{z}$; with Gaussian $\mathbf{z}$ it is the gradient of the Gaussian-smoothed loss [Nesterov and Spokoiny, 2017], with convex [Duchi et al., 2015] and bandit [Flaxman et al., 2005] complexity theory (survey: Liu et al., 2020), analog-VLSI origins [Jabri and Flower, 1992, Cauwenberghs, 1992, Flower and Jabri, 1992], evolution strategies [Wierstra et al., 2014, Hansen et al., 2003, Hansen, 2016] that compete with policy gradients [Salimans et al., 2017, Mania et al., 2018], relate to SGD [Zhang et al., 2017] and apply to supervised learning [Lenc et al., 2019], adaptive-moment updates [Chen et al., 2019] and directional-derivative learning [Silver et al., 2022]. MeZO [Malladi et al., 2023] fine-tunes billion-parameter language models with two-point SPSA in place; 1.5-SPSA [Chaubard, 2025b] adds a per-direction three-point curvature estimate to damp the step in sharp regions. DeepZero [Chen et al., 2024] and Zhang et al. [2024] benchmark estimators on one large model; we vary the estimator *and* the architecture at a scale where every method runs to its one-hour limit, and our SPSA, 1.5-SPSA and RS are the two-point, curvature-preconditioned and best-of-$K$ instances of this family.

**Node perturbation.** Node perturbation injects noise into the units and correlates the loss change with it, a local update [Widrow and Lehr, 1990, Werfel et al., 2003, 2005, Fiete and Seung, 2006] whose stability and weight diffusion Hiratani et al. [2022] analysed; Züge et al. [2023] found weight perturbation often similar or better on temporally extended tasks, revisited here. Dalm et al. [2026] recast it as a directional derivative (*iterative* NP), proposed *activity-based* NP from the measured activity change, and added the input decorrelation of Ahmad et al. [2023], which also accelerates backpropagation [Dalm et al., 2024, Ahmad, 2024]; 1.5-ANP [Chaubard, 2025a] adds a three-point curvature preconditioner to central-difference activity perturbation; we implement all six.

**Local rules and benchmarks.** Forward-forward [Hinton, 2022], our representative local rule, replaces the backward pass by two forward passes with positive and negative data and a local "goodness" objective per layer. Global-error broadcasting [Clark et al., 2021], feedback alignment

[Lillicrap et al., 2016, Nøkland, 2016] and synthetic gradients [Jaderberg et al., 2017] still need a learned or random backward signal, and forward gradients [Baydin et al., 2022, Ren et al., 2023] a Jacobian-vector product, so they are outside the zero-order scope. Controlled comparisons exist for the feedback-alignment family [Bartunov et al., 2018, Launay et al., 2020, Sanfiz and Akrout, 2021]; we are not aware of one that puts weight perturbation, the node-perturbation family and forward-forward on the same models under one budget (unabridged: Appendix D).

# 3 Methods

## 3.1 Problem setup and the shared update rule

Let $\boldsymbol{\theta} \in \mathbb{R}^P$ be the parameters of a network with $L$ perturbable layers and $\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N}\sum_{n=1}^{N} \ell_n(\boldsymbol{\theta})$ the mini-batch cross-entropy loss, where $\ell_n$ is the per-example loss (for language modelling, the per-sequence mean of the $T = 64$ next-character cross-entropies). Every method writes an estimate $\hat{\mathbf{g}} \approx \nabla_{\boldsymbol{\theta}}\mathcal{L}$ into the gradient slot of every parameter—backpropagation the exact gradient, the zero-order methods their estimator, forward-forward the gradient of its local objectives—and the same optimizer consumes it: Adam [Kingma and Ba, 2015] with $(\beta_1, \beta_2) = (0.9, 0.99)$, learning rate $\eta$ and no schedule (Eq. (1), Appendix A). Every probe method has a second scalar, the probe scale $\sigma$: the standard deviation of the parameter perturbation for the weight-space methods and of the node noise for the node-perturbation family; all probes are drawn from $\mathcal{N}(\mathbf{0}, \mathbf{I})$ and scaled by $\sigma$. The protocol *ties* the two, $\eta = \sigma = \varepsilon$, so that one scalar is swept per method; Section 5.4 measures what the tie costs. Because Adam normalises $\hat{\mathbf{g}}$ by a running estimate of its own magnitude, Eq. (1) is approximately invariant to a constant rescaling of the estimator: the normalisation constants of the estimators affect the first few hundred steps and the relative weighting of estimates across steps, not the asymptotic step size, so the comparison is insensitive to the different conventions in the literature.

## 3.2 Optimizers

Appendix A has full definitions (Eqs. (2)–(9)) and the budget (Table 3); here, estimator and cost.

**SPSA.** With $K = 8$ antithetic pairs of Gaussian probes $\mathbf{z}_k$, $\hat{\mathbf{g}}_{\text{SPSA}}$ averages the central differences $[\mathcal{L}(\boldsymbol{\theta} + \sigma\mathbf{z}_k) - \mathcal{L}(\boldsymbol{\theta} - \sigma\mathbf{z}_k)]\,\mathbf{z}_k/2\sigma$ (Eq. (2)): the Gaussian-smoothing estimator of Nesterov and Spokoiny [2017] and multi-direction SPSA [Spall, 1992] with Gaussian rather than Rademacher directions, perturbed in place as in MeZO [Malladi et al., 2023]; 16 passes, no clean pass.

**1.5-SPSA.** 1.5-SPSA [Chaubard, 2025b] adds a clean pass (17 passes) and divides each central difference by $c_k = \max(\hat{\kappa}_k^{\alpha}, \lambda)$, $\hat{\kappa}_k$ the three-point second difference along $\mathbf{z}_k$ ($\alpha = 0.1$, $\lambda = 1$; Eq. (3)). Under Adam $c_k$ is inert (Section 5.6), so Section 5.5 also runs the published recipe (Rademacher directions, plain SGD, $\eta = \sigma$); we use Gaussian directions throughout.

**Random search.** Random search [Rastrigin, 1963, Matyas, 1965, Schumer and Steiglitz, 1968] evaluates the clean loss and $K = 15$ candidates $\boldsymbol{\theta} + \sigma\mathbf{z}_k$ and keeps the best, expressed for the shared update rule as a pseudo-gradient along the improving direction and zero when none improves (Eq. (4)); through Adam this only approximates the direct jump (Appendix A).

**Node perturbation (NP).** Classic NP [Widrow and Lehr, 1990, Werfel et al., 2003, Fiete and Seung, 2006, Hiratani et al., 2022] injects Gaussian noise $\sigma\boldsymbol{\xi}_l$ into *every* layer's pre-activations on a noisy pass and estimates layer $l$'s gradient as the outer product of the noise-weighted per-example loss change $\delta\mathcal{L}_n\boldsymbol{\xi}_{l,n}/\sigma$ with the clean layer input (Eq. (5)); one clean and 15 noisy passes, averaged.

**Activity-based NP (ANP).** ANP [Dalm et al., 2026] replaces the injected noise by the measured activity difference $\delta\mathbf{a}_l = \tilde{\mathbf{a}}_l - \mathbf{a}_l$, normalised by the squared activity perturbation summed over all layers and scaled by the total number $c$ of perturbed units (Eq. (6)); same 1+15 passes.

**1.5-ANP.** 1.5-ANP [Chaubard, 2025a] applies 1.5-SPSA's curvature preconditioner to central-difference activity perturbation ($K = 8$ pairs plus a clean pass, 17 passes), per *attribution unit* (an image example or a text token), with $\alpha = 0.5$ and $\lambda = 10^{-3}$ (Eq. (7)); $\alpha = 0$ gives central-difference ANP, the control that separates attribution from preconditioning (own recipe: Section 5.5).

**Iterative NP (INP).** INP [Dalm et al., 2026] perturbs *one* layer per pass, so the loss change is attributable to it: a directional derivative along the noise, $\lfloor 15/L \rfloor$ draws per layer (Eq. (8)).

**Decorrelated ANP (DANP).** DANP [Dalm et al., 2026] is ANP on layer inputs decorrelated by a matrix $\mathbf{R}_l$ trained by the local rule of Ahmad et al. [2023] after each Adam step (Eq. (9)).

**Forward-forward (FF).** FF [Hinton, 2022] trains each block with a local goodness objective (mean of squares and threshold $\theta_{\text{FF}} = 2$ of the re-implementation of Pezeshki [2023]) on positive and negative data, forwarded as one batch with the neutral-label inputs (three $N$-sized passes per step on the image tasks, two on the text tasks), and never propagates an error between blocks; we report Hinton's label accumulation and a linear readout on the top block (Appendix A).

**Backpropagation (reference) and forward-pass budget.** BP feeds the exact gradient of one forward and one backward pass to the same Adam update. The probe methods thus spend 16 forward passes per step, the two 1.5-order rules 17, FF three or two and BP one forward plus one backward (Table 3), so steps per hour differ between methods (Appendix F), and the node-perturbation family obtains one loss difference *per example* whereas SPSA and RS obtain one batch-mean loss per pass: an asymmetry intrinsic to node perturbation that the equal-pass budget compares.

## 4 Experimental Protocol

**Tasks and architectures.** An MLP on MNIST [LeCun et al., 1998], a ConvNet on CIFAR-10 [Krizhevsky, 2009], and a character-level LSTM [Hochreiter and Schmidhuber, 1997] and Transformer [Vaswani et al., 2017] on Penn Treebank [Marcus et al., 1993, Mikolov, 2012] ($V = 50$ symbols; 5.10M / 0.40M / 0.45M characters in train / valid / test; a word-level embedding would by itself exceed the parameter budget), all with ≈40k parameters (Table 4, Appendix B). Weights are Xavier-uniform [Glorot and Bengio, 2010] with zero biases (LSTM forget-gate bias 1), and the Transformer's layer normalisation [Ba et al., 2016] has no affine parameters, so every parameter lives in a perturbable layer; the last 5k training images are validation, inputs are standardised, batches are 128 images or 64 sequences of 64 characters, and there is no augmentation, dropout or weight decay.

**Stage 1: $\varepsilon$ screening.** Every (task, method) pair is run once at $\varepsilon \in \{0.1, 0.01, 0.001\}$ ($\eta = \sigma = \varepsilon$, fixed throughout) for one hour of training on one NVIDIA RTX 5090, with validation and test metrics at initialisation, every 5 minutes of training and at the end (evaluation time excluded); the two 1.5-order rules were added later under the stage-2 settings. Each pair selects the $\varepsilon$ with the lowest validation loss at any evaluation (FF on the image tasks: the highest label-accumulation validation accuracy). Because 26 of the 32 original pairs selected this grid's lower edge, $\varepsilon = 10^{-4}$ was added for every pair under the stage-2 settings, and because that value won for eight methods on both image tasks, $\varepsilon = 10^{-5}$ for those sixteen pairs: a selected value must have a worse neighbour on each side (176 screening runs, Appendix E). On the sequence tasks $10^{-4}$ loses to $10^{-3}$ for every rule on the LSTM and for five of the ten on the Transformer, so most of their selections stay at the stage-1 values (Table 8).

**Stage 2: replication.** The selected configuration of every pair is run with three seeds (initialisation, data order and probes), evaluated every **minute** of training time, in full fp32 arithmetic (TF32 disabled; Section 5.6 measures its effect), with checkpoints at the best validation loss. A run is stopped and marked diverged when a loss is non-finite or when its clean validation loss exceeds ten times its initial value at two consecutive evaluations, which stops collapsed networks and not over-confident but accurate ones (Appendix B: the replaced stage-1 rule and pairs whose stage-1 criteria disagree). For BP, which has no probe, the reference on each task is the learning rate and weight decay with the lowest validation loss among $\eta \in \{10^{-2}, 10^{-3}, 10^{-4}\}$ with and without decoupled weight decay $10^{-2}$ [Loshchilov and Hutter, 2019], replicated with three seeds.

**Reporting.** The primary metric is the *best validation checkpoint*: per pair and seed, the evaluation with the lowest validation loss over the hour, whose test loss and accuracy (classification) or bits per character (loss/ $\ln 2$) are reported as the seed mean with the sample standard deviation in units of the last digit. The end-of-hour value measures stability; FF on the image tasks is selected and reported

Table 1: Main result: test loss and test accuracy (%) or bits per character at the *best validation checkpoint* of the selected $\varepsilon$, mean over three seeds with the standard deviation in units of the last digit in parentheses (s1: single stage-1 run). Best probe method per task (by mean loss) in bold; BP is the reference and FF a local (non-probe) rule reported by label-accumulation accuracy [readout accuracy in brackets].

| | MNIST / MLP | | CIFAR-10 / ConvNet | | PTB / char-LSTM | | PTB / char-Transformer | |
|---|---|---|---|---|---|---|---|---|
| Method | loss | acc. (%) | loss | acc. (%) | loss | bits/char | loss | bits/char |
| BP | 0.102 | 97.1 | 0.907(10) | 69.2(6) | $1.231^{s1}$ | 1.78 | $1.260^{s1}$ | 1.82 |
| SPSA | 0.142(5) | 95.8(2) | 1.200(51) | 57.8(20) | $2.048^{s1}$ | 2.96 | $2.181^{s1}$ | 3.15 |
| 1.5-SPSA | 0.140(8) | 95.9(2) | 1.215(48) | 56.7(19) | 2.054(8) | 2.96(1) | 2.155(26) | 3.11(4) |
| RS | 0.168(12) | 94.9(3) | 1.292(56) | 53.7(22) | $2.124^{s1}$ | 3.06 | $2.279^{s1}$ | 3.29 |
| NP | 0.107(2) | 96.9(0) | 1.081 | 62.0 | $2.015^{s1}$ | 2.91 | 2.336 | 3.37 |
| ANP | 0.110 | 96.6 | **1.060** | **63.1** | $2.969^{s1}$ | 4.28 | 2.720 | 3.92 |
| 1.5-ANP | 0.159(7) | 95.7(4) | 1.491 | 47.3 | 2.167(21) | 3.13(3) | 2.589 | 3.74 |
| INP | **0.107(4)** | **96.9(2)** | 1.072 | 63.2 | $\mathbf{1.955}^{s1}$ | **2.82** | $\mathbf{2.151}^{s1}$ | **3.10** |
| DANP | 0.211(2) | 94.0(1) | 1.094 | 64.1 | $2.608^{s1}$ | 3.76 | $2.513^{s1}$ | 3.63 |
| FF | 2.252(34) | 93.1(3) [14.4] | 2.032 | 33.6 [26.4] | $2.658^{s1}$ | 3.84 | 2.921 | 4.21 |

by label-accumulation accuracy (readout accuracy in brackets); mean ranks are over the eight probe methods (95 % bootstrap interval over seeds); test data are never used for selection (Appendix B).

**Ablations.** (i) *Untied probe scale and step size*: the $3 \times 3$ grid $\eta, \sigma \in \{10^{-2}, 10^{-3}, 10^{-4}\}$ plus $\sigma = 0.1$ at $\eta = 10^{-4}$ per probe method and task, screened for 30 minutes, with the best untied setting replicated for an hour with three seeds where it beats the best tied value by at least 3 % at the 30-minute checkpoint. (ii) *The 1.5-order rules under their own recipe*: plain SGD, fp64 losses, their own $(\eta, \sigma)$ grids, $\alpha = 0$ controls and the LSTM state damping (Section 5.5). (iii) *Controls under the protocol*: 1.5-ANP at the selected step size ($\eta = 10^{-4}$, Adam) with $\sigma \in \{10^{-2}, 10^{-3}\}$ decoupled, fp64 losses and $\alpha \in \{0, 0.25, 0.5, 1\}$ (Table 2), 1.5-SPSA with a non-inert factor. (iv) *Estimator alignment*: the cosine between every estimator and the true gradient of a fixed batch against the number of forward passes (Section 5.6). (v) *Resources*: steps per hour, seconds per step and peak GPU memory of every run (Appendix F). Appendix B specifies each in full and lists hardware, software, code commits and logging.

# 5 Results

## 5.1 Main comparison

Table 1 reports the best-validation-checkpoint test loss and accuracy or bits per character of every method on every task, Table 11 the same runs at the end of the hour and Table 8 the screening behind every selection (Appendix F: minute of the best checkpoint and seeds). On the image tasks the node-perturbation family leads and three of its members tie: INP and NP share the lowest loss on MNIST (0.107, 96.9 %) with ANP 0.003 behind, and ANP, INP and NP are within 0.02 of each other on CIFAR-10 (1.060–1.081, 62–63 %), where DANP has the highest accuracy (64.1 %) at a slightly higher loss. The weight-space rules follow at a distance that grows with the task: SPSA and 1.5-SPSA reach 95.8–95.9 % on MNIST and 57–58 % on CIFAR-10, RS 94.9 and 53.7 %. On the sequence tasks INP is the best probe method (2.82 and 3.10 bits per character), 1.5-SPSA and SPSA are second on the Transformer (3.11 and 3.15) and NP second on the LSTM (2.91), and ANP is last on both (4.28 and 3.92). Across the four tasks INP has the best mean rank among the eight probe methods at the best checkpoint (1.25, bootstrap interval 1.25–2.0), ahead of NP (3.0), SPSA and 1.5-SPSA (4.0), ANP (5.0), RS (5.75), DANP (6.25) and 1.5-ANP (6.75). Against backpropagation on the same one-minute grid (97.1 % on MNIST and 69.2 % on CIFAR-10 at its own best learning rate, Section 5.3) the best probe rule is 0.2 points behind on MNIST and 5–6 points behind on CIFAR-10, and on the LSTM and the Transformer the gap is 1.0 and 1.3 bits per character, a factor 1.6–1.7 in test loss. The end-of-hour view (Table 11) promotes the weight-space rules, which drift least, and leaves the sequence tasks almost unchanged.

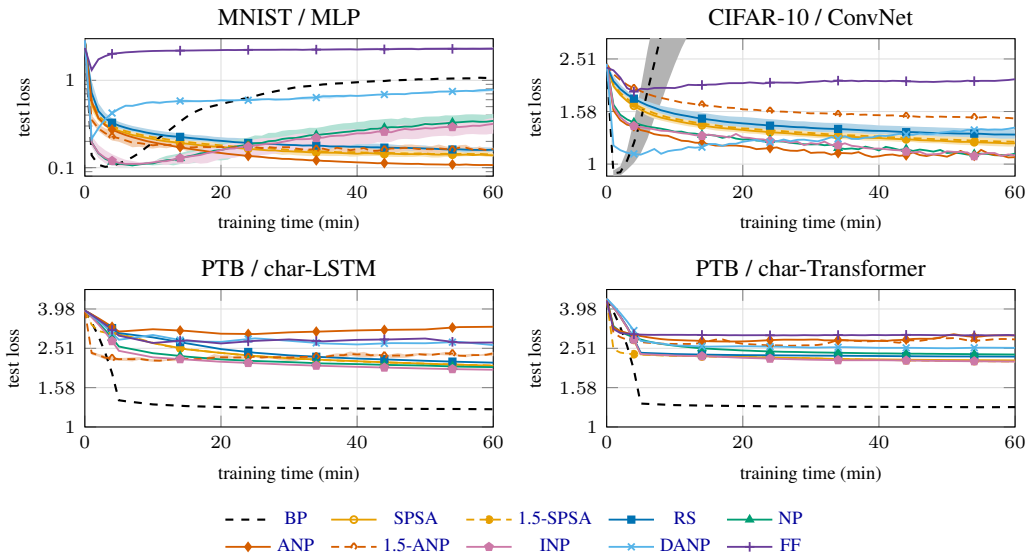



Figure 1: Test loss versus training time (minutes; evaluation excluded) of the selected runs: seed mean (line) and seed range (shaded). Log $y$-axis. The 1.5-order rules are dashed in their parent's colour.

### 5.2 Training curves and equal-compute view

Figure 1 shows test loss against training time for the selected runs (seed mean and range); Figure 3 (Appendix C) shows the same runs against the number of forward passes, which removes implementation- and host-dependent throughput from the comparison. The curves have two shapes. On the image tasks BP reaches its best validation loss within the first two minutes and then grows over-confident for the rest of the hour (97.1 % throughout on MNIST; 0.91 to 11 in test loss on CIFAR-10 at 66–69 %), whereas at $\varepsilon = 10^{-4}$ every probe rule is still improving or flat at minute 60: the weight-space rules and 1.5-SPSA have their best checkpoint in the last five minutes on both tasks, ANP and INP on CIFAR-10 as well, and only NP and 1.5-ANP on MNIST and DANP peak early and drift (Appendix F). On both sequence tasks every curve descends monotonically and is still descending at minute 60, roughly one bit per character above BP. Per forward pass the node rules lead on the image tasks and trail on the LSTM (Appendix C).

### 5.3 The backpropagation reference

With Adam and no schedule, backpropagation completes 4.3M steps on MNIST and 2.0M on CIFAR-10 in the hour and overfits within minutes: at $\eta = 10^{-3}$ its CIFAR-10 validation loss never returns below its initial value, and at the selected reference ($\eta = 10^{-5}$ on MNIST, $10^{-4}$ on CIFAR-10) the best checkpoint sits at minute 1–3 (97.1 and 69.2 %) and the hour ends at 97.1 and 65.6 % (Table 6). On the sequence tasks $\eta = 10^{-3}$ is best, BP is still improving at minute 60 (1.78 and 1.82 bits per character) and the reference is unproblematic. [AdamW rows of Table 6 land at 12:30 UTC.]

### 5.4 Probe scale and step size

The tied screening (Table 8, Figure 4) was decisive but for the wrong knob: $\varepsilon = 0.1$ collapses most probe runs and BP itself on both image tasks, $\varepsilon = 0.01$ still loses eight probe runs, $10^{-3}$ loses none, and the lower values added after the review improve eight of the ten methods on both image tasks once more—SPSA from 0.293 to 0.132 in validation loss on MNIST and from 1.62 to 1.27 on CIFAR-10 at $10^{-4}$—while $10^{-5}$ is worse for all but three pairs (BP and ANP on MNIST, 1.5-ANP on CIFAR-10, where it ties or wins by a margin within seed noise). The untied grid (Table 7, Appendix C) attributes this to the step size alone: at fixed $\eta = 10^{-4}$ the probe scale changes the 30-minute loss by at most 0.03 on MNIST and 0.2 on CIFAR-10 for every rule except 1.5-ANP, whereas on the image tasks $\eta = 10^{-2}$ is catastrophic for every rule at every $\sigma$. Adam makes each

Table 2: 1.5-ANP under the protocol's optimizer at the step size the $\varepsilon$ screening selected on the image tasks (Adam, $\eta = 10^{-4}$), with the probe scale decoupled from the step size and the losses evaluated in fp64: test loss (accuracy or bits per character) at the best validation checkpoint, seed 0. The $\alpha = 0$ columns remove the curvature weighting; the "damped" row is the LSTM with the recurrent-state damping (0.5) of its specification. The last two columns are the tied-protocol selections of 1.5-ANP and ANP (mean over seeds).

| Task | $\sigma = 10^{-2}$ | $\sigma = 10^{-2}, \alpha = 0$ | $\sigma = 10^{-3}$ | $\sigma = 10^{-3}, \alpha = 0$ | 1.5-ANP tied | ANP tied |
|---|---|---|---|---|---|---|
| MNIST / MLP | 0.132 (96.2) | 0.115 (96.8) | 0.133 (96.2) | 0.112 (97.0) | 0.159(7) | 0.110 |
| CIFAR-10 / ConvNet | 1.279 (56.9) | 1.222 (56.5) | 1.307 (54.7) | 1.120 (60.4) | 1.491 | 1.060 |
| PTB / char-LSTM | – | – | – | – | 2.167(21) | 2.969 |
| PTB / char-LSTM, damped | – | – | – | – | – | – |
| PTB / char-Transformer | – | – | – | – | 2.589 | 2.720 |

step about $\eta$ per coordinate whatever the estimate's scale, so a noisy estimate wants a small step and many of them [Dalm et al., 2026, Malladi et al., 2023]. On the sequence tasks $\eta = 10^{-4}$ loses to $10^{-3}$ for every rule on the LSTM (the recurrent model completes only 20–35k steps per hour) and for five of ten on the Transformer. The best untied cell beats the best tied one by more than 3 % only for 1.5-ANP on both image tasks (24 and 17 %, at $\sigma \geq 10^{-3}$), DANP on the Transformer (6 %) and ANP on MNIST (4 %); the tied protocol is therefore not what holds the other rules back.

### 5.5 The 1.5-order rules under their own recipe

Table 5 (Appendix C) runs both rules as published—plain SGD, fp64 losses, their own $(\eta, \sigma)$ and probe-count grids—and Table 2 runs 1.5-ANP under the protocol's optimizer at the step size the screening selected, with its probe scale freed. Neither recipe beats the protocol: 1.5-SPSA with SGD and 64 antithetic pairs per step reaches 0.176 on MNIST and 1.336 on CIFAR-10 where its own Adam run stands at 0.160 and 1.313 after the same 30 minutes, and its $\alpha = 0$ control (plain SPSA with SGD) is within 0.01 or better on every task (0.162, 1.344, 2.845 versus 2.955, 2.259 versus 2.258). The curvature factor of 1.5-SPSA is inert under both optimizers: with $\alpha = 0.1$ and $\lambda = 1$ it varies by 5 % across directions and steps. For 1.5-ANP the picture is the same but more informative. Under Adam at $\eta = 10^{-4}$ with fp64 losses it improves from 0.159 to 0.132 on MNIST and from 1.49 to 1.28 on CIFAR-10 (56.9 %) at $\sigma = 10^{-2}$, i.e. its tied selection was limited by a probe scale too small for a second difference in fp32, but its $\alpha = 0$ control is better still (0.112 and 1.120, 97.0 and 60.4 %) and comes within 0.002 and 0.06 of ANP (0.110, 1.060); $\alpha = 0.25$ lies in between and $\alpha = 1$ is harmful (0.204 and 1.98). Under its own SGD recipe the $\alpha = 0$ control again equals or beats the curvature-weighted rule on every task. The ingredient that makes 1.5-ANP usable on the sequence models, where it beats ANP by 1.15 and 0.18 bits per character under the protocol, is thus not the curvature exponent but the rest of its estimator: the per-unit normalisation by the measured activity difference and the token-local credit assignment. [LSTM and Transformer rows of Table 2 (with and without damping) land at 12:30 UTC.]

### 5.6 Estimator quality

Figure 2 measures, on one fixed batch per task, the cosine between each estimator and the true gradient against the number of forward passes spent on it, after a 2000-step backpropagation warm-up, in fp32 on the GPU (Appendices C and F). On the MLP the node-perturbation rules obtain from the protocol's 16 passes a direction with cosine 0.4–0.6 to the gradient, rising to 0.95–0.96 for NP and INP and 0.81–0.83 for ANP and DANP with 1024 passes, whereas the weight-space rules get 0.01 from 16 passes and 0.07–0.11 from 1024, the $\sqrt{K/P}$ scaling of a 40k-dimensional random search. On the ConvNet every estimator is below 0.1 at 16 passes: its 26,682 perturbed unit-positions approach the number of weights, so the node estimate needs thousands of probes before it carries signal. On the LSTM and the Transformer no estimator exceeds 0.26 and 0.17 at 1024 passes (INP reaches 0.40 and 0.33 at 4096): perturbations that propagate through shared weights over 64 positions decorrelate the loss difference from any single unit, and 1.5-ANP, DANP and INP are the least affected. 1.5-SPSA is indistinguishable from SPSA at every probe count, and 1.5-ANP's per-unit factor lowers the alignment of ANP from 0.81 to 0.69 on the MLP and from 0.52 to 0.27 on the ConvNet at $\sigma = 10^{-3}$ while raising it on the sequence models (0.26 versus 0.04 on the LSTM). Down to $\sigma = 10^{-5}$ the first-difference estimators keep their alignment in fp32, whereas 1.5-ANP's second difference reaches the rounding

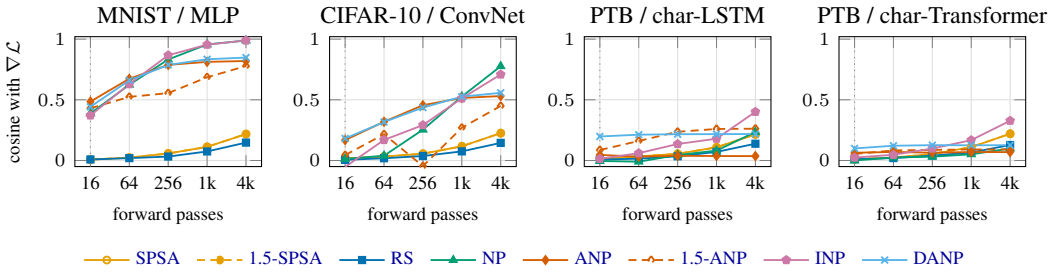



Figure 2: Cosine between each estimator and the true gradient of one batch (whole parameter vector) versus the number of forward passes spent, $\sigma = 10^{-3}$, after a 2000-step backpropagation warm-up. The protocol spends 16–17 passes per step.

floor (0.74 at $10^{-2}$ to 0.46 at $10^{-5}$ on the MLP), which is why it needs $\sigma \geq 10^{-3}$ (Section 5.5); TF32 convolutions lower the ConvNet cosines of NP and INP at $\sigma = 10^{-3}$ (Table 17), so stage 2 ran in fp32.

### 5.7 Analysis

**Which zero-order method wins?** INP, if a single rule must be chosen: the best probe method on both sequence models, tied for best on MNIST and within 0.01 of the best on CIFAR-10, and the best mean rank at the best checkpoint and at the end of the hour. ANP, NP and INP are interchangeable on the image tasks and DANP is the fastest there; on the sequence tasks the order is INP, then the weight-space rules and NP, then the curvature-weighted and decorrelated rules, with ANP last. The gap to backpropagation is 0.2 points on MNIST, 5–6 points on CIFAR-10 and 1.0–1.3 bits per character on the sequence tasks, where every probe curve is still descending.

**Weight probes versus node probes.** SPSA and RS perturb all $P \approx 40$k weights and obtain one scalar loss difference per probe, so their 16 passes yield a direction with cosine 0.01 to the gradient on the MLP (Figure 2), whereas the node rules obtain a loss difference per example and assign credit per unit (cosine 0.4–0.6); at the right step size the gap is 1 point on MNIST and 5 points on CIFAR-10 at the best checkpoint, and it closes on the sequence models, where SPSA ends within 0.14 and 0.05 bits per character of INP and ahead of NP on the Transformer, as Züge et al. [2023] found for temporally extended tasks. The effective-dimension argument—node probes win where the number of perturbed units is small relative to $P$ (90 on the MLP) and lose their advantage where it is comparable (26,682 on the ConvNet, 25,728 and 48,256 on the sequence models)—fits the sequence-task convergence but not the order within the node family, where ANP beats NP by one point on CIFAR-10 with the same unit count and loses to it by 1.4 bits per character on the LSTM.

**Curvature preconditioning.** 1.5-SPSA reproduces SPSA on every task to within 0.03 in loss (0.140 versus 0.142 on MNIST, 1.215 versus 1.200 on CIFAR-10, 2.96 versus 2.96 and 3.11 versus 3.15 bits per character) at the price of the clean pass (13–15 % fewer steps per hour on the image tasks, none on the sequence models where the clean pass is amortised): with $\alpha = 0.1$ and $\lambda = 1$ the per-direction factor is nearly constant and Adam removes the remaining common scalar, and Section 5.5 shows the same under plain SGD. 1.5-ANP is one point behind ANP on MNIST and 16 points behind on CIFAR-10 under the tied protocol, because a second difference at $\sigma \leq 10^{-4}$ sits at the fp32 rounding floor; with $\sigma = 10^{-2}$ and fp64 losses it recovers to within 0.02 of ANP on MNIST, and its $\alpha = 0$ control matches ANP on both tasks. On the sequence models its token-local attribution repairs ANP's failure (3.13 versus 4.28 bits per character on the LSTM, 3.74 versus 3.92 on the Transformer) without reaching INP or SPSA; per step it is 7 % slower than ANP on the MLP and 15–50 % faster on the other three models (Table 14).

**Activity-based, iterative and decorrelated variants.** The measured activity difference (ANP) is worth one point over NP on CIFAR-10 and nothing on MNIST at the right step size, less than the 12 points the stage-1 grid suggested, but on the recurrent and attention models its normalisation by a squared activity difference summed over 25k–48k unit-positions, whose perturbations propagate through shared weights, produces updates of erratic size (the spikes in Figure 1) and the worst probe

result on both tasks. INP's one-layer-at-a-time attribution has only $\lfloor 15/L \rfloor$ draws per layer, yet removes all cross-layer noise from each estimate (the highest 1024-pass cosine of any rule on the MLP) and is never worse than third on any task. Decorrelation helps only where Dalm et al. [2026] reported it: DANP is the fastest rule on CIFAR-10 (best checkpoint at minute 8, the highest accuracy of any probe rule, 64.1 %) but drifts to 1.38 by minute 60, is behind on MNIST (94.0 %), remains far behind on the sequence tasks (3.76 and 3.63 bits per character) and was the screening's most fragile rule (NaN within 21 steps at $\varepsilon = 0.1$ on both image models).

**Sequence models and forward-forward.** Against count models on the training text (unigram 4.32 bits/char, bigram 3.31, trigram 2.61; Appendix G) the zero-order rules learn roughly bigram statistics in an hour—the best four sit between the bigram and the trigram on the LSTM and barely past the bigram on the Transformer—and BP (1.78 and 1.82) a real character model; every probe curve is still descending at minute 60, so part of the gap is budget rather than estimator, but at 17–35k steps per hour on the LSTM closing it would take days. FF, not a probe method, trains fast (1.7M steps per hour on the MLP) but its two inference rules disagree: on MNIST the linear readout peaks early and collapses to 14 %, whereas label accumulation on the same blocks gives 93.1 %; on CIFAR-10 both numbers are 26–34 %, and on the sequence tasks FF is second-worst on the LSTM and worst on the Transformer (3.84 and 4.21 bits per character; Appendix C).

## 6 Discussion and Limitations

**What the final-loss protocol measures.** Under a fixed Adam step with no schedule every image-task run peaks within minutes and then drifts: Adam normalises each coordinate, so once the estimate's signal is exhausted the parameters perform a random walk of step $\eta$ and the network becomes over-confident while its accuracy barely moves. The loss after a fixed hour therefore ranks methods by where their random walk settles, and selecting $\varepsilon$ by it produced degenerate choices in stage 1; hence stage 2 reports the best validation checkpoint as primary and the final value for stability. A learning-rate schedule, not in the protocol, would be the other remedy.

**Screening, budget and scale.** The selection rests on one seed per $\varepsilon$, so the error bars are conditional on $\varepsilon^*$ and exclude selection noise. The budget is small and identical for all methods, which favours methods that make progress quickly; the models are small by modern standards, and because the variance of weight-space estimators grows with $P$, the standing of SPSA and RS against the node-perturbation family may change at larger scale. Wall-clock comparisons reward implementations that batch probes well: the probe passes of this code are a Python loop and all runs executed eight at a time on one host, so steps per hour (Appendix F) depend on the host and the equal-pass view (Figure 3) is the implementation-independent comparison.

**Faithfulness.** SPSA and 1.5-SPSA use Gaussian rather than Rademacher directions, RS is expressed through Adam, 1.5-ANP applies the recurrent-state damping of its specification only in the controls of Table 2, FF follows the reference re-implementation at the parameter budget, and DANP's decorrelation rate is fixed and un-normalised (Appendix C). Character-level results are not comparable to word-level perplexities.

## 7 Conclusion

We compared nine backprop-free training rules and backpropagation on four ≈40k-parameter architectures under one equal-budget protocol. The screening sweep's first result is methodological: with Adam the step size, not the probe scale, decides how far a noisy estimator gets in an hour, and at $\eta = 10^{-4}$ every probe rule improves and the weight-space rules close most of their gap on the image tasks. If one zero-order rule must be chosen it is iterative node perturbation, best on both character-level models and tied for best on the image tasks; classic and activity-based node perturbation match it on images and the activity-based rule fails on sequences; weight-space perturbation competes where the number of perturbed units approaches the number of weights; curvature preconditioning adds nothing to SPSA under Adam or SGD, and the sequence-model gains of 1.5-ANP over ANP survive setting its curvature exponent to zero. Per forward pass the node rules keep up with backpropagation on the image tasks; per unit of time the best probe rule is 0.2 and 6 points behind, and on the language models every zero-order rule is still about one bit per character behind after an hour, every curve still

descending. The protocol, code, job lists and all run logs are public; adding a rule to the grid costs about 25 GPU-hours.

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# A Optimizer definitions

This appendix gives the shared update rule and the full definition of every rule summarised in Section 3.2; Table 3 lists the forward-pass budget of every method per step.

**Shared update rule.** Every estimate $\hat{\mathbf{g}}$ is consumed by Adam [Kingma and Ba, 2015] with $(\beta_1, \beta_2) = (0.9, 0.99)$, learning rate $\eta$ and no schedule:

$$\mathbf{m} \leftarrow \beta_1 \mathbf{m} + (1-\beta_1)\,\hat{\mathbf{g}}, \qquad \mathbf{v} \leftarrow \beta_2 \mathbf{v} + (1-\beta_2)\,\hat{\mathbf{g}}^2, \qquad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\,\frac{\hat{\mathbf{m}}}{\sqrt{\hat{\mathbf{v}}} + 10^{-8}}. \tag{1}$$

**Adam's constants.** $(\beta_1, \beta_2) = (0.9, 0.99)$ is a protocol choice made before any run (the usual $\beta_2 = 0.999$ would average the second moment over a thousand steps, several seconds of these runs); it is shared by every method and is not tuned.

**SPSA (central difference, Gaussian probes).** With $K = 8$ antithetic pairs of probes $\mathbf{z}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_P)$,

$$\hat{\mathbf{g}}_{\text{SPSA}} = \frac{1}{K}\sum_{k=1}^{K} \frac{\mathcal{L}(\boldsymbol{\theta} + \sigma\mathbf{z}_k) - \mathcal{L}(\boldsymbol{\theta} - \sigma\mathbf{z}_k)}{2\sigma}\,\mathbf{z}_k, \tag{2}$$

the Gaussian-smoothing estimator of Nesterov and Spokoiny [2017] and the multi-direction form of SPSA [Spall, 1992] with Gaussian rather than Rademacher directions. Each pair costs two forward passes on the batch-mean loss, so the 16-pass budget buys eight directions. Parameters are perturbed in place and restored exactly from a flat copy, as in MeZO [Malladi et al., 2023]; no clean pass is spent, and the training loss logged for SPSA is the mean of the 16 probe losses.

**1.5-SPSA.** 1.5-SPSA [Chaubard, 2025b] spends one clean pass on $\mathcal{L}_0 = \mathcal{L}(\boldsymbol{\theta})$ and uses the three-point second difference along every direction to damp the step where the loss is sharp:

$$\begin{aligned} \hat{\mathbf{g}}_{\text{1.5-SPSA}} &= \frac{1}{K}\sum_{k=1}^{K} \frac{\mathcal{L}(\boldsymbol{\theta} + \sigma\mathbf{z}_k) - \mathcal{L}(\boldsymbol{\theta} - \sigma\mathbf{z}_k)}{2\sigma\, c_k}\,\mathbf{z}_k, \\ c_k &= \max\!\left(\hat{\kappa}_k^{\alpha}, \lambda\right), \qquad \hat{\kappa}_k = \frac{\left|\mathcal{L}(\boldsymbol{\theta} + \sigma\mathbf{z}_k) - 2\mathcal{L}_0 + \mathcal{L}(\boldsymbol{\theta} - \sigma\mathbf{z}_k)\right|}{\sigma^2}, \end{aligned} \tag{3}$$

with the saturating exponent $\alpha = 0.1$ and floor $\lambda = 1$ of the published code. With $K = 8$ pairs the step costs 17 passes. The published method uses Rademacher directions and a plain SGD step $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\,\hat{\mathbf{g}}$ with $\eta = \sigma$ tied, i.e. $\Delta\boldsymbol{\theta} = -\frac{1}{2K}\sum_k \frac{\Delta\mathcal{L}_k}{c_k}\mathbf{z}_k$, many directions per step and few steps [Chaubard, 2025b]. Under the protocol the estimate goes to Adam like every other method, which makes $c_k$ inert (Section 5.6); Section 5.5 therefore also runs the rule under its own recipe. We use Gaussian directions, like SPSA, throughout.

**Random search.** Random search [Rastrigin, 1963, Matyas, 1965, Schumer and Steiglitz, 1968] evaluates the clean loss $\mathcal{L}_0$ and $K = 15$ candidates $\boldsymbol{\theta} + \sigma \mathbf{z}_k$ and keeps the best, $k^* = \arg\min_k \mathcal{L}(\boldsymbol{\theta} + \sigma \mathbf{z}_k)$. To fit the shared update rule it is expressed through the pseudo-gradient

$$\hat{\mathbf{g}}_{\text{RS}} = \begin{cases} \dfrac{\mathcal{L}_{k^*} - \mathcal{L}_0}{\sigma}\, \mathbf{z}_{k^*} & \text{if } \mathcal{L}_{k^*} < \mathcal{L}_0, \\ \mathbf{0} & \text{otherwise,} \end{cases} \tag{4}$$

which points along the improving direction and is zero when no candidate improves. Through Adam this is only an approximation of the jump $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \sigma \mathbf{z}_{k^*}$: the first step moves by $\approx \eta\, \mathrm{sign}(\mathbf{z}_{k^*})$ per coordinate, later steps are bent by momentum, and rejected steps still move the parameters along the decaying momentum; the direct-jump variant ("jumping" random search without Adam) is implemented in the released code but was not run within the budget.

**Node perturbation (NP).** Classic node perturbation [Widrow and Lehr, 1990, Werfel et al., 2003, Fiete and Seung, 2006, Hiratani et al., 2022] perturbs the units rather than the weights. For layer $l$ with input $\mathbf{h}_{l-1}$, pre-activation $\mathbf{a}_l = \mathbf{W}_l \mathbf{h}_{l-1} + \mathbf{b}_l$ and post-activation $\mathbf{h}_l = f(\mathbf{a}_l)$, a noisy pass injects Gaussian noise into *every* layer and propagates it, $\tilde{\mathbf{a}}_l = \mathbf{W}_l \tilde{\mathbf{h}}_{l-1} + \mathbf{b}_l + \sigma \boldsymbol{\xi}_l$ with $\boldsymbol{\xi}_l \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Writing $\delta\mathcal{L}_n = \tilde{\ell}_n - \ell_n$ for the change of the per-example loss, the estimator for layer $l$ is

$$\hat{\mathbf{g}}^{\text{NP}}_{\mathbf{W}_l} = \frac{1}{N} \sum_{n=1}^{N} \frac{\delta\mathcal{L}_n}{\sigma}\, \boldsymbol{\xi}_{l,n}\, \mathbf{h}_{l-1,n}^{\top}, \qquad \hat{\mathbf{g}}^{\text{NP}}_{\mathbf{b}_l} = \frac{1}{N} \sum_{n=1}^{N} \frac{\delta\mathcal{L}_n}{\sigma}\, \boldsymbol{\xi}_{l,n}, \tag{5}$$

the outer product of the noise-weighted loss change with the *clean* layer input (for convolutions, the matching correlation over unfolded patches; for embeddings, the one-hot input), averaged over 15 noisy passes. $\delta\mathcal{L}_n \boldsymbol{\xi}_l / \sigma$ equals $\delta\mathcal{L}_n \mathbf{v}_l / \sigma^2$ with $\mathbf{v}_l = \sigma \boldsymbol{\xi}_l$, the form of Dalm et al. [2026]. For a layer applied several times per forward (the LSTM gate layer, once per time step) the same $\delta\mathcal{L}_n$ weights every call.

**Activity-based node perturbation (ANP).** Because injected noise propagates, the actual perturbation of layer $l$ is the measured activity difference $\delta\mathbf{a}_l = \tilde{\mathbf{a}}_l - \mathbf{a}_l$ rather than $\sigma\boldsymbol{\xi}_l$. Activity-based NP [Dalm et al., 2026] uses this difference and normalises by the norm of the total activity perturbation across all layers:

$$\hat{\mathbf{g}}^{\text{ANP}}_{\mathbf{W}_l} = \frac{1}{N} \sum_{n=1}^{N} c\, \frac{\delta\mathcal{L}_n}{\|\delta\mathbf{a}_n\|^2}\, \delta\mathbf{a}_{l,n}\, \mathbf{h}_{l-1,n}^{\top}, \qquad \|\delta\mathbf{a}_n\|^2 = \sum_{l'=1}^{L} \|\delta\mathbf{a}_{l',n}\|^2, \tag{6}$$

with $c = \sum_l N_l$ the total number of perturbed units (all layers, all time steps or positions; the input is not counted), following the authors' reference implementation. For a single perturbed unit it reduces to Eq. (5).

**1.5-ANP.** 1.5-ANP [Chaubard, 2025a] combines activity-based perturbation with antithetic pairs and the three-point curvature preconditioner of 1.5-SPSA, applied per *attribution unit* $u$: an example on the image tasks and a (sequence, position) token on the language-modelling tasks ("token-local" attribution, with the LSTM gate layer called once per position). Each of $K = 8$ pairs injects $\pm\sigma\boldsymbol{\xi}$ into every layer; with $\mathbf{d}_u = \mathbf{a}_u^+ - \mathbf{a}_u^-$ the realized central activity difference over the $N_u$ perturbed nodes of $u$ and $\ell_u^0, \ell_u^{\pm}$ the clean and perturbed losses of $u$,

$$\hat{c}_u = \frac{4N_u\, |\ell_u^+ - 2\ell_u^0 + \ell_u^-|}{\|\mathbf{d}_u\|^2}, \qquad \beta_u = \frac{N_u\, (\ell_u^+ - \ell_u^-)}{\|\mathbf{d}_u\|^2\, (\hat{c}_u + \lambda)^{\alpha}}, \qquad \hat{\mathbf{g}}^{\text{1.5-ANP}}_{\mathbf{a}_u} = \frac{1}{K} \sum_{k=1}^{K} \beta_u^{(k)} \mathbf{d}_u^{(k)}, \tag{7}$$

with $\lambda = 10^{-3}$ ($10^{-4}$ on the LSTM in the specification) and $\alpha = 0.5$; the weight update is the batch mean of $\hat{\mathbf{g}}_{\mathbf{a}_u} \mathbf{h}_u^{\top}$ as in Eq. (6), and $N_u$ counts the perturbed nodes of one unit (90 on the MLP, 402 per LSTM time step, 754 per Transformer token). With $\alpha = 0$ the rule is central-difference ANP, the control that separates attribution from preconditioning. The step costs 17 passes. The specification uses plain SGD with a learning rate 20–50 times $\sigma$ and damps the deviation of the perturbed recurrent state by one half at every time step; the protocol runs ties $\eta = \sigma$ under Adam without damping, and Section 5.5 runs the rule under its own recipe with both. The three-point second differences that define $\hat{c}_u$ are, at $\sigma = 10^{-3}$, of the order of the fp32 rounding of the loss, so the own-recipe runs evaluate the losses in fp64 (Section 5.6).

Table 3: Forward-pass budget per step. $L$ is the number of perturbable layers (3 for the MLP and the LSTM, 5 for the ConvNet, 15 for the Transformer).

| Method | Clean | Probe passes | Backward | Estimate per step |
|---|---|---|---|---|
| BP | 1 | 0 | full | exact gradient |
| SPSA | 0 | 16 | – | 8 central differences, averaged |
| 1.5-SPSA | 1 | 16 | – | 8 curvature-scaled central differences |
| RS | 1 | 15 | – | best of 15 candidates |
| NP, ANP, DANP | 1 | 15 (all layers) | – | 15 noise draws, averaged (DANP: $+\mathbf{R}_l$ update) |
| 1.5-ANP | 1 | 16 (all layers, 8 pairs) | – | 8 curvature-scaled central differences per unit |
| INP | 1 | 15 (one layer each) | – | $\lfloor 15/L \rfloor$ draws per layer |
| FF | 3 / 2 | 0 | per block | local goodness gradients |

**Iterative node perturbation (INP).** Iterative NP [Dalm et al., 2026] perturbs *one* layer per noisy pass, so that the loss change $\delta\mathcal{L}_n^{(l)}$ is attributable to that layer and the estimator is a directional derivative along the injected noise $\mathbf{v}_{l,n} = \sigma\boldsymbol{\xi}_{l,n}$,

$$\hat{\mathbf{g}}_{\mathbf{W}_l}^{\text{INP}} = \frac{1}{N}\sum_{n=1}^{N} N_l \,\frac{\delta\mathcal{L}_n^{(l)}}{\|\mathbf{v}_{l,n}\|^2}\,\mathbf{v}_{l,n}\,\mathbf{h}_{l-1,n}^{\top}, \tag{8}$$

with $N_l$ the number of perturbed units of layer $l$. The 15 noisy passes give $\lfloor 15/L \rfloor$ draws per layer: 5 for the MLP and the LSTM ($L = 3$), 3 for the ConvNet ($L = 5$) and 1 for the Transformer ($L = 15$). For the LSTM gate layer, called once per time step, the norm in Eq. (8) is taken over the injected noise of all calls, whereas the reference code of Dalm et al. [2026] (feed-forward only) uses the measured activity difference; the two coincide for a layer called once.

**Decorrelated ANP (DANP).** Decorrelated ANP [Dalm et al., 2026] is ANP in which every layer input is first decorrelated by a square matrix $\mathbf{R}_l$ that is not a parameter of the model, $\hat{\mathbf{h}}_{l-1} = \mathbf{R}_l\mathbf{h}_{l-1}$, with $\mathbf{R}_l$ initialised to the identity and trained by the local rule of Ahmad et al. [2023],

$$\mathbf{R}_l \leftarrow \mathbf{R}_l - \eta_R\left(\hat{\mathbf{C}}_l - \operatorname{diag}\hat{\mathbf{C}}_l\right)\mathbf{R}_l, \qquad \hat{\mathbf{C}}_l = \tfrac{1}{M}\textstyle\sum_{m=1}^{M}\hat{\mathbf{h}}_{l-1,m}\hat{\mathbf{h}}_{l-1,m}^{\top}, \tag{9}$$

from the decorrelated inputs recorded on the clean pass ($M$ rows: examples, or examples $\times$ positions / patches). The decorrelation rate is fixed at $\eta_R = 10^{-3}$, the value of Dalm et al. [2026] and the default of their code, and is not tied to $\varepsilon$; $\mathbf{R}_l$ is updated once per step after the Adam step. Embeddings, whose inputs are one-hot, are not decorrelated.

**Forward-forward (FF).** Forward-forward [Hinton, 2022] trains each block with a local objective and never propagates an error between blocks. With goodness $G = \operatorname{mean}_j h_{l,j}^2$ (the mean of squares and threshold $\theta_{\text{FF}} = 2$ of the reference re-implementation of Pezeshki [2023]; Hinton uses the sum of squares), the local loss on a positive/negative pair is $\ell_{\text{FF}} = \log(1 + e^{\theta_{\text{FF}} - G_{\text{pos}}}) + \log(1 + e^{G_{\text{neg}} - \theta_{\text{FF}}})$, and activities are $\ell_2$-normalised before the next block. A block is one hidden layer of the MLP, one convolution–ReLU–pool stage or the dense layer of the ConvNet, the LSTM layer, and one pre-LN block of the Transformer; for text the goodness is computed per position. For images, positive data carry the correct label overlaid on the first 10 pixels and negative data a wrong label; for text, negatives have 30 % of their characters replaced at random. The positive, negative and (for images) neutral-label inputs are forwarded as one batch, i.e. three $N$-sized passes per step on the image tasks and two on the text tasks. Two inference rules are reported: Hinton's label accumulation (the label whose overlay yields the highest total goodness over all blocks but the first, Section 3.3 of Hinton [2022]; we also log the all-blocks variant) and a linear readout trained on the normalised, detached top-block activity of the neutral-label pass (Hinton's softmax reads all hidden layers but the first; reading only the top block keeps the readout inside the parameter budget). The LSTM block's output lies in $(-1, 1)$, so its goodness cannot exceed 1; that block uses $\theta_{\text{FF}} = 0.25$.

**Backpropagation (reference).** One forward and one backward pass produce the exact gradient, fed to the same Adam update. It is the reference, not a zero-order method.

## B Protocol details

Table 4 lists the four tasks of Section 4; the reporting conventions, the divergence rule, the ablations and the hardware and software of the protocol are specified in full below.

Table 4: Tasks, splits (train / validation / test) and architectures. Hidden nonlinearities are ReLU; convolutions are $3\times3$ with $2\times2$ max-pooling; the LSTM [Hochreiter and Schmidhuber, 1997] has one layer of 80 units; the Transformer [Vaswani et al., 2017] has two pre-LN blocks of width 48, 4 heads, feed-forward width 64, learned positional embeddings and context $T = 64$. *Units* is the number of pre-activation values one node-perturbation probe perturbs for a single example, summed over layers, time steps and spatial positions (the Transformer's position embedding, $64 \times 48$ values, is perturbed once per batch and counted once); it is the normalisation constant $c$ of Eq. (6).

| Key | Data | Split | Model | Params | Units |
|---|---|---|---|---|---|
| `mlp_mnist` | MNIST | 55k / 5k / 10k | MLP 784–48–32–10 | 39,578 | 90 |
| `conv_cifar10` | CIFAR-10 | 45k / 5k / 10k | conv16–conv32–conv32–fc48–10 | 39,450 | 26,682 |
| `lstm_ptb` | PTB (char) | standard | emb32–LSTM80–$V$ | 41,810 | 25,728 |
| `transformer_ptb` | PTB (char) | standard | emb48 + pos64–2 blocks–$V$ | 39,250 | 48,256 |

**Reporting conventions.** The primary metric is the *best validation checkpoint*: for each pair and seed, the evaluation with the lowest validation loss over the hour, whose test loss, test accuracy (classification) or bits per character (loss/ $\ln 2$, language modelling) is reported as the mean over seeds with the standard deviation in units of the last digit ("0.140(5)" is $0.140 \pm 0.005$; the sample standard deviation over three seeds; cells without parentheses are single runs). The value at the end of the hour is reported as a secondary stability metric, and for the two classification tasks the checkpoint with the highest validation *accuracy* is reported as well (Appendix F), since loss and accuracy drift apart under a fixed step. FF on the image tasks is selected and reported by label-accumulation accuracy, with the readout accuracy in brackets. Mean ranks are over the eight probe methods, with a 95 % bootstrap interval over seeds. Test data are never used for selection.

**Selection with disagreeing criteria.** Where the stage-1 selection by best and by final validation loss disagree in a non-degenerate way (DANP on both image tasks), both $\varepsilon$ are replicated and the better mean validation loss decides.

**Divergence rule.** Stage 2 stops a run and marks it diverged when a loss is non-finite or when its clean validation loss exceeds ten times its initial value at two consecutive evaluations. Stage 1 used a factor two and additionally compared each method's own training loss with its first-step value; the former stopped backpropagation on CIFAR-10 at 67 % accuracy and the latter was method-dependent because SPSA's training loss is a probe mean, so both were replaced; a run that the old rule did not stop is identical under the new one.

**Ablations.** (i) *Untied probe scale and step size*: for every probe method and task the full grid $\eta \in \{10^{-2}, 10^{-3}, 10^{-4}\} \times \sigma \in \{10^{-2}, 10^{-3}, 10^{-4}\}$ plus $\sigma = 0.1$ at $\eta = 10^{-4}$, screened for 30 minutes at 1-minute granularity (the tied diagonal is taken from the one-hour runs at minute 30); the best untied setting of a pair is then replicated for one hour with three seeds where it beats the best tied diagonal cell by at least 3 % (at most twelve pairs, largest gains first), so that every rule is also compared at its own best $(\eta, \sigma)$ at equal compute and without any schedule. (ii) *The 1.5-order rules under their own recipe*: plain SGD instead of Adam, fp64 loss evaluation, 1.5-SPSA with $\eta = \sigma \in \{10^{-2}, 3 \cdot 10^{-3}, 10^{-3}, 3 \cdot 10^{-4}, 10^{-4}\}$ and 8 or 64 pairs with the $\alpha = 0$ (1SPSA) control, 1.5-ANP with $\eta \in \{0.05, 0.02, 0.01, 0.005\}$ decoupled from $\sigma \in \{10^{-2}, 10^{-3}\}$ with the $\alpha = 0$ control, a $\lambda = 10^{-2}$ floor and the recurrent-state damping on the LSTM; 30-minute screening, then three seeds for an hour at the selected setting (Section 5.5). (iii) *Controls under the protocol*: 1.5-ANP with Adam at the step size the screening selected on the image tasks ($\eta = 10^{-4}$), the probe scale decoupled ($\sigma \in \{10^{-2}, 10^{-3}\}$), fp64 losses, the curvature exponent $\alpha \in \{0, 0.25, 0.5, 1\}$ and, on the LSTM, the recurrent-state damping (Table 2); 1.5-ANP with $\alpha = 0$ and with fp64 losses at $\varepsilon = 10^{-3}$; 1.5-SPSA with a non-inert factor ($\alpha = 0.5$, $\lambda = 10^{-3}$). (iv) *Estimator alignment*: the cosine between every estimator and the true gradient of a fixed batch as a function of the number of forward passes, per layer, at initialisation and after a 2000-step backpropagation warm-up, in fp32, TF32 and fp64, for $\sigma$ from $10^{-1}$ to $10^{-5}$. (v) *Resources*: every run records its steps, seconds of training time per step and peak GPU memory (Appendix F); throughput is measured with eight runs sharing one host.

**Hardware and software.** All runs use one RTX 5090 per run on an 8-GPU vast.ai node (8 runs at a time; PyTorch 2.9.1, CUDA 12.8), with the code at `https://github.com/Fchaubard/`

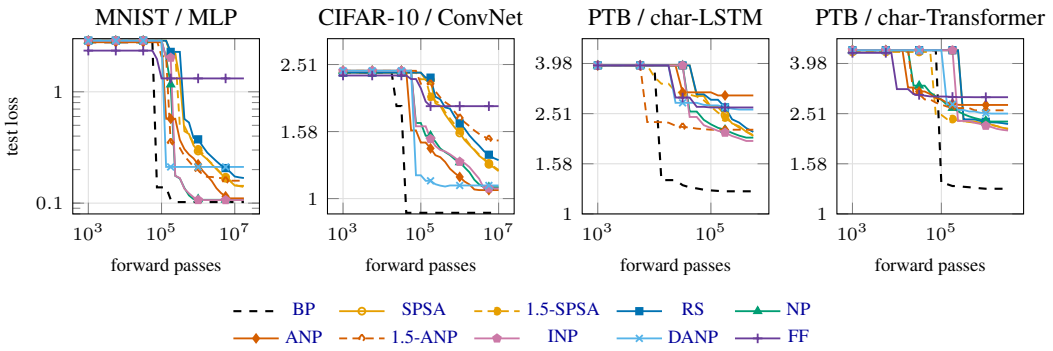



Figure 3: Equal-compute view: test loss versus the number of forward passes (one BP step counts one pass plus its backward), interpolated between the 1-minute evaluations, mean over seeds.

`zero-order-ablations` (stage 1: commits `db100bc` and `f5424e9`; stage 2: `97c03f5`). Every run is logged to Weights & Biases (project `zoo-survey`) and writes a JSON file with its configuration, evaluation history and final metrics.

## C Additional results and discussion

Table 5: The 1.5-order rules under their own recipe (plain SGD, fp64 losses, 30-minute screening): the best configuration by validation loss and its best-checkpoint test loss (accuracy or bits per character), the $\alpha = 0$ control at the same grid, and the protocol run (Adam, tied $\varepsilon$) at minute 30.

| Task | Rule | best config | test loss (acc / bpc) | $\alpha = 0$ control | protocol (Adam) @30 min | 1SPSA / ANP-SGD |
|---|---|---|---|---|---|---|
| MNIST / MLP | 1.5-SPSA | lr0.003 K129 l64 sgd | 0.176 (94.8) | 0.162 (94.9) | 0.160(16) | – |
| | 1.5-ANP | lr0.01 sig0.01 lam0.001 l64 sgd | 0.137 (96.0) | 0.121 (96.6) | 0.167(15) | – |
| CIFAR-10 / ConvNet | 1.5-SPSA | lr0.003 K129 l64 sgd | 1.336 (51.7) | 1.344 (51.6) | 1.313(50) | – |
| | 1.5-ANP | lr0.005 sig0.001 lam0.001 l64 sgd | 2.356 (20.5) | 2.406 (9.4) | 1.559 | – |
| PTB / char-LSTM | 1.5-SPSA | lr0.01 K17 l64 sgd | 2.955 (4.26) | 2.845 (4.10) | 2.234(1) | – |
| | 1.5-ANP | lr0.05 sig0.01 lam0.0001 l64 sgd | 2.463 (3.55) | 2.424 (3.50) | 2.167(21) | – |
| PTB / char-Transformer | 1.5-SPSA | lr0.01 K17 l64 sgd | 2.258 (3.26) | 2.259 (3.26) | 2.230(4) | – |
| | 1.5-ANP | lr0.005 sig0.01 lam0.001 l64 sgd | 2.763 (3.99) | 2.763 (3.99) | 2.589 | – |

**Equal-pass view.** Figure 3 shows the selected runs of Figure 1 against the number of forward passes, which removes the implementation- and host-dependent throughput of each method from the comparison. Per forward pass (Figure 3) the picture favours the node rules on the image tasks: after $10^5$ passes DANP is at 0.365 on MNIST and 1.149 on CIFAR-10 against 0.480 and 1.315 for BP, and after $10^6$ passes ANP, INP and NP sit at 0.13–0.16 on MNIST while BP, which has by then overfitted, is at 2.46; on CIFAR-10 at $10^6$ passes DANP and ANP (1.19, 1.21) are ahead of BP (1.48). On the LSTM the order is reversed: after $10^5$ passes BP is at 1.24 against 2.20–2.27 for INP, 1.5-ANP and NP, because a backward pass through 64 time steps buys far more than a perturbed forward pass does.

**Backpropagation reference.** With Adam at $\eta = 10^{-3}$ and no schedule, backpropagation completes 4.2M steps on MNIST and 2.4M on CIFAR-10 in the hour and overfits the 40k-parameter models within the first minute: on MNIST its validation loss is lowest at the first evaluation (97.2 and 97.0 % test accuracy for two seeds, test loss 0.37–0.40) and ends at 97.4 % with test loss 2.4–2.6; on CIFAR-10 the $\eta = 10^{-3}$ run has already passed its best validation loss by minute 1 (67.7 % accuracy with a validation loss above its initial value) and is stopped by the divergence rule, so the selected run is $\eta = 10^{-2}$, 62.7 % at minute 1 and 59.3 % at the end; four stage-1 runs with 30-second evaluations saw 69.7 % on CIFAR-10 and 97.5 % on MNIST after 30 seconds, i.e. the best point of BP on these models lies between the first two evaluations of the protocol. On the sequence tasks BP is still improving at minute 60 (1.78 and 1.82 bits per character) and the reference is unproblematic.

**Probe scale and step size.** The stage-1 sweep (Appendix E, Figure 4) is decisive about the tied scalar: at $\varepsilon = 0.1$, 15 of the 24 runs of the six original probe methods diverge (all four DANP runs,

Table 6: Backpropagation reference under the stage-2 protocol: test accuracy (%) or bits per character at the best validation checkpoint (with its minute) and at the end of the hour, for learning rates $\eta$ and decoupled weight decay (AdamW). The $\eta = 0.001$ row without decay is the three-seed reference of Table 1; the $\eta = 0.01$ row is the stage-1 run. A dagger marks a run stopped by the divergence rule.

| | | MNIST | | CIFAR-10 | | LSTM | | Transformer | |
|---|---|---|---|---|---|---|---|---|---|
| lr | decay | best (min) | final | best (min) | final | best (min) | final | best (min) | final |
| 0.01 | 0 | 12.7 (0) | 67.1$^\dagger$ | 62.7 (1) | 59.3 | 1.85 (20) | 1.93 | 1.89 (40) | 1.90 |
| 0.001 | 0 | 97.2(2) (1) | 97.4(0) | 9.4 (0) | 66.9$^\dagger$ | 1.78 (60) | 1.78 | 1.82 (55) | 1.82 |
| 0.0001 | 0 | 97.2(1) (1) | 97.1(0) | 69.2(6) (1) | 65.6(3) | 2.12 (60) | 2.12 | 1.93 (60) | 1.93 |
| 0.001 | 0.01 | – | – | – | – | – | – | – | – |
| 0.0001 | 0.01 | – | – | – | – | – | – | – | – |

Table 7: Probe scale decoupled from the step size: test loss of the best validation checkpoint within 30 minutes for every probe method and task (one seed; the tied $\eta = \sigma = 10^{-3}$ column is the three-seed mean of the stage-2 runs at minute 30 and the tied $10^{-2}$ column the stage-1 run). A dagger marks a diverged run.

| | | $\eta = 10^{-2}$ | | | $\eta = 10^{-3}$ | | | $\eta = 10^{-4}$ | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Task | Method | $\sigma$=$10^{-2}$ | $10^{-3}$ | $10^{-4}$ | $\sigma$=$10^{-2}$ | $10^{-3}$ | $10^{-4}$ | $\sigma$=$10^{-2}$ | $10^{-3}$ | $10^{-4}$ |
| MNIST / MLP | SPSA | 2.229 | – | 2.001 | – | 0.365 | – | 0.165 | 0.173 | **0.154** |
| | 1.5-SPSA | 2.226$^\dagger$ | 2.159 | 1.977 | 0.332 | 0.365 | 0.392 | 0.171 | 0.180 | **0.160** |
| | RS | 2.761 | 2.496 | 2.308 | – | 0.440 | – | 0.206 | 0.208 | **0.182** |
| | NP | 0.984 | – | 0.503 | – | 0.143 | – | 0.110 | 0.112 | **0.107** |
| | ANP | 0.557 | – | 0.496 | – | 0.126 | – | 0.109 | **0.107** | 0.112 |
| | 1.5-ANP | 1.574 | 1.987 | 2.101 | 0.154 | 0.218 | 0.324 | **0.127** | 0.146 | 0.167 |
| | INP | 0.606$^\dagger$ | – | 0.370 | – | 0.134 | – | 0.110 | 0.115 | **0.107** |
| | DANP | 0.317 | 0.299 | 0.303 | – | **0.211** | – | 0.216 | 0.215 | 0.215 |
| CIFAR-10 / ConvNet | SPSA | 3.234$^\dagger$ | – | 2.303 | – | 1.645 | – | 1.304 | 1.353 | **1.292** |
| | 1.5-SPSA | 2.367$^\dagger$ | 5.402 | 2.993 | 1.525 | 1.609 | 1.604 | 1.322 | 1.373 | **1.313** |
| | RS | 2.328$^\dagger$ | 2.406$^\dagger$ | 9.260$^\dagger$ | – | 1.730 | – | 1.463 | 1.447 | **1.373** |
| | NP | 2.406$^\dagger$ | – | 2.331$^\dagger$ | – | 1.407 | – | 1.192 | **1.188** | 1.190 |
| | ANP | 2.312 | – | 2.306 | – | 1.175 | – | 1.262 | 1.161 | **1.103** |
| | 1.5-ANP | 2.304 | 2.304 | 2.306 | **1.360** | 1.775 | 1.860 | 1.428 | 1.473 | 1.639 |
| | INP | 2.304 | – | 2.303 | – | 1.315 | – | **1.159** | 1.165 | 1.181 |
| | DANP | 1.150 | 1.170 | 1.174 | – | **1.094** | – | 1.444 | 1.258 | 1.238 |
| PTB / char-LSTM | SPSA | **2.109** | – | 2.129 | – | 2.236 | – | 3.100 | 3.090 | 3.086 |
| | 1.5-SPSA | **2.095** | 2.150 | 2.127 | 2.217 | 2.234 | 2.224 | 3.102 | 3.114 | 3.107 |
| | RS | 2.306 | **2.250** | 2.264 | – | 2.314 | – | 3.253 | 3.270 | 3.272 |
| | NP | 2.218 | – | 2.205 | – | **2.138** | – | 2.949 | 2.946 | 2.987 |
| | ANP | 3.356$^\dagger$ | – | 2.999 | – | **2.969** | – | 3.008 | 3.007 | 3.008 |
| | 1.5-ANP | 2.167 | **2.139** | 2.139 | 2.236 | 2.268 | 2.235 | 2.628 | 2.718 | 2.693 |
| | INP | **2.023** | – | 2.033 | – | 2.073 | – | 2.837 | 2.826 | 2.816 |
| | DANP | 3.132$^\dagger$ | 2.758 | 2.857 | – | **2.709** | – | 3.059 | 3.051 | 3.059 |
| PTB / char-Transformer | SPSA | 2.536 | – | 2.580 | – | **2.237** | – | 2.348 | 2.352 | 2.352 |
| | 1.5-SPSA | 2.547 | 2.584 | 2.570 | **2.209** | 2.230 | 2.225 | 2.358 | 2.360 | 2.359 |
| | RS | 2.736 | 2.792 | 2.768 | – | **2.306** | – | 2.721 | 2.401 | 2.400 |
| | NP | 2.964 | – | 2.937 | – | **2.372** | – | 2.397 | 2.399 | 2.397 |
| | ANP | 4.496$^\dagger$ | – | 2.827 | – | 3.065 | – | 2.936 | **2.719** | 2.720 |
| | 1.5-ANP | 2.703 | 2.651 | 2.694 | **2.581** | 2.593 | 2.633 | 2.675 | 2.671 | 2.589 |
| | INP | 2.605 | – | 2.660 | – | **2.198** | – | 2.329 | 2.324 | 2.332 |
| | DANP | 2.533 | 2.414 | **2.384** | – | 2.525 | – | 3.770 | 3.066 | 3.101 |

three each of NP, ANP and RS, two of SPSA, none of INP), BP itself collapses on both image tasks at this step size, and the surviving probe runs make little progress; at $\varepsilon = 0.01$ seven probe runs diverge, five of them node-perturbation runs, and BP is stopped on MNIST; at $\varepsilon = 0.001$ no probe run diverges and 25 of the 32 original pairs select this value. The screening of the two 1.5-order rules under the stage-2 settings follows the same pattern: six of their sixteen runs at $\varepsilon \geq 0.01$ diverge (1.5-SPSA on both image tasks at 0.1 and 0.01, 1.5-ANP on CIFAR-10 and the LSTM at 0.1), and every pair selects $\varepsilon = 0.001$ except 1.5-ANP on the LSTM (0.01). Because $\varepsilon$ is both the Adam step and the probe scale, these counts do not say whether a method diverged from noise or from step size, and the published rules do not obey the tie (Dalm et al. [2026] fix $\sigma = 10^{-3}$ with learning rates down to $10^{-5}$; MeZO uses $\sigma = 10^{-3}$ with learning rates of $10^{-6}$–$10^{-7}$). Table 7 separates the two.

**Estimator alignment.** Figure 2 measures, on one fixed batch per task, the cosine between each estimator and the true gradient as a function of the number of forward passes spent on it, after a 2000-step backpropagation warm-up (Appendix F gives the per-layer values and the same measurement at initialisation). On the MLP the node-perturbation rules obtain, from the protocol's 16 passes, a direction with cosine 0.40–0.60 to the gradient (ANP 0.60, INP 0.47, DANP 0.41, NP 0.40, 1.5-ANP 0.29), rising to 0.96–0.97 for NP and INP and 0.83 for ANP and DANP with 1024 passes, whereas the weight-space rules get 0.01 from 16 passes and 0.07–0.12 from 1024, the $\sqrt{K/P}$ scaling of a 40k-dimensional random search. On the ConvNet at initialisation every estimator is below 0.1 at 16 passes, and with 1024 passes ANP reaches 0.58, INP 0.40, NP and DANP 0.32–0.33, SPSA 0.12 and 1.5-ANP 0.09: the convolutional layers, whose 26,682 perturbed unit-positions approach the number of weights, need thousands of probes before the node estimate carries signal, which is why NP and INP train the ConvNet slowly and the weight-space rules hardly at all. The curvature preconditioners behave as their results suggest: 1.5-SPSA is indistinguishable from SPSA at every probe count, because with $\alpha = 0.1$ and $\lambda = 1$ its per-direction factor lies between 1 and 1.3, i.e. a nearly constant per-step scalar (0.53–0.62 on the MLP, varying by 5 % across steps) that Adam's normalisation removes, while 1.5-ANP's per-example factor with $\alpha = 0.5$ lowers the alignment of central-difference ANP from 0.83 to 0.62 on the MLP at $\sigma = 10^{-3}$ (and to 0.09 on the ConvNet), where the fp32 second differences that define it are at the rounding floor. Numerical precision matters at the protocol's smallest probe scale: with TF32 convolutions, which PyTorch enables by default, the ConvNet cosine of NP at $\sigma = 10^{-3}$ and 1024 passes drops from 0.445 to 0.420 and that of INP from 0.425 to 0.358 (fp64: 0.475 and 0.484), while SPSA and ANP are unaffected and nothing changes at $\sigma = 10^{-2}$ (Table 17). Stage 1 ran with TF32 on; stage 2 runs in full fp32.

**Sequence models.** On the test split a uniform model scores 5.64 bits/char, a unigram 4.32, a bigram 3.31 and an add-0.1 trigram 2.61 (count models estimated on the training text; Appendix G). On the LSTM, INP, NP, SPSA, 1.5-SPSA and RS (2.82–3.06) sit between the bigram and the trigram and 1.5-ANP (3.09) just below the bigram; DANP (3.76) and FF (3.83) are worse than the bigram and ANP (4.28) is near the unigram. On the Transformer only INP, 1.5-SPSA, SPSA and RS (3.10–3.29) edge past the bigram. BP reaches 1.78 and 1.82 bits per character, well beyond the trigram: after one hour the zero-order rules have learned roughly bigram statistics and backpropagation a real character model. Every probe curve is still descending at minute 60, so part of the gap is budget rather than estimator, but at 17k–35k steps per hour on the LSTM closing it would take days.

**Forward-forward.** FF is not a probe method and trains fast (1.7M steps per hour on the MLP), but its two inference rules disagree. The linear readout on the neutral-label features of the 32-unit top block peaks within the first minute on MNIST (87.5 % at $\varepsilon = 0.001$) and then collapses to 14–20 % as its training cross-entropy rises towards $\ln 10$; Hinton's label accumulation on the same trained blocks gives 93.4 % when summed over all blocks but the first (the protocol's rule, which on this two-block MLP means the second block alone) and 96.0 % when summed over both blocks, at every $\varepsilon$. On CIFAR-10 all three numbers are 26–34 %, and on the sequence tasks FF with corrupted-sequence negatives is the second-worst rule (3.83 and 4.22 bits per character): its goodness separates real from corrupted text, but the normalised features support only a weak next-character readout.

**Faithfulness.** SPSA and 1.5-SPSA use Gaussian rather than Rademacher directions and RS is expressed through Adam (the direct-jump variant is implemented but was not run); 1.5-ANP omits the recurrent-state damping of its specification; FF's goodness, threshold and readout follow the reference re-implementation and the parameter budget rather than Hinton's 2,000-unit layers; DANP's decorrelation rate is fixed and un-normalised, which is why it reaches NaN within 21 steps at $\varepsilon = 0.1$ on both image models (a single Adam step inflates the ReLU activities by an order of magnitude and the $\mathbf{R}_l$ update then grows tenfold per step). All of these are stated where the method is defined.

## D Extended related work

This appendix is the unabridged version of Section 2.

**Weight-space perturbation.** Simultaneous perturbation stochastic approximation [Spall, 1992, 1998] estimates a gradient from two loss evaluations at $\boldsymbol{\theta} \pm \sigma\mathbf{z}$ with a random direction $\mathbf{z}$; with Gaussian $\mathbf{z}$ the estimator is the gradient of the Gaussian-smoothed loss [Nesterov and Spokoiny,

2017], with complexity understood in the convex [Duchi et al., 2015] and bandit [Flaxman et al., 2005] settings (see Liu et al. [2020] for a survey). Weight perturbation was proposed for analog VLSI training by Jabri and Flower [1992], with stochastic error descent [Cauwenberghs, 1992] and summed-weight perturbation [Flower and Jabri, 1992] as early variance reductions. Evolution strategies use the same estimator with many directions, fitness shaping and natural-gradient corrections [Wierstra et al., 2014, Hansen et al., 2003, Hansen, 2016]; Salimans et al. [2017] and Mania et al. [2018] showed them competitive with policy-gradient reinforcement learning, Zhang et al. [2017] related them to SGD, and Lenc et al. [2019] applied them to supervised learning. Adaptive-moment updates on top of zero-order estimates were studied by Chen et al. [2019], and Silver et al. [2022] analysed learning from directional derivatives along random directions. MeZO [Malladi et al., 2023] fine-tunes billion-parameter language models with two-point SPSA and in-place perturbation; 1.5-SPSA [Chaubard, 2025b] adds a three-point curvature estimate along each direction to damp the step in sharp regions. DeepZero [Chen et al., 2024] and Zhang et al. [2024] benchmark zero-order estimators on one large model; we vary the estimator *and* the architecture at a scale where every method can be run to its one-hour limit. Our SPSA, 1.5-SPSA and RS baselines are the two-point, curvature-preconditioned and best-of-$K$ instances of this family.

**Node perturbation.** Node perturbation injects noise into the units of the network and correlates the resulting loss change with the noise, which yields a weight update local to each layer [Widrow and Lehr, 1990, Werfel et al., 2003, 2005, Fiete and Seung, 2006]. Hiratani et al. [2022] analysed its stability and showed that it suffers from weight diffusion in mismatched networks; Züge et al. [2023] compared weight and node perturbation on temporally extended tasks and found weight perturbation often similar or better, a finding our sequence tasks revisit. Dalm et al. [2026] reformulated node perturbation as a directional derivative (*iterative* NP), proposed *activity-based* NP that uses the measured activity change rather than the injected noise, and combined it with the input decorrelation rule of Ahmad et al. [2023], which also accelerates backpropagation [Dalm et al., 2024, Ahmad, 2024]. 1.5-ANP [Chaubard, 2025a] applies a three-point curvature preconditioner to central-difference activity perturbation. We implement all six variants.

**Local rules and benchmarks.** The forward-forward algorithm [Hinton, 2022] replaces the backward pass by two forward passes with positive and negative data and trains each layer to separate them with a local "goodness" objective. Rules that broadcast a global error vector [Clark et al., 2021], feedback alignment [Lillicrap et al., 2016, Nøkland, 2016] and synthetic gradients [Jaderberg et al., 2017] still require a learned or random backward signal, and forward gradients [Baydin et al., 2022, Ren et al., 2023] require a Jacobian-vector product through the network; they are outside the zero-order scope of this benchmark, which includes forward-forward as the representative local rule. Controlled comparisons of biologically motivated rules exist for the feedback-alignment family [Bartunov et al., 2018, Launay et al., 2020, Sanfiz and Akrout, 2021]; we are not aware of one that puts weight perturbation, the node-perturbation family and forward-forward on the same models under one budget.

## E Stage-1 screening grid

Table 9 lists the final validation and test loss of every stage-1 run of the eight original methods, Figure 4 the final test loss against $\varepsilon$, and Figure 5 every run's test-loss curve. The validation and test values of a collapsed run can differ by orders of magnitude at the same evaluation because the loss of a collapsed network is dominated by a few examples with huge logits. The screening of 1.5-SPSA and 1.5-ANP under the stage-2 settings (1-minute evaluations, fp32, 1 h) selected $\varepsilon = 0.001$ for every pair except 1.5-ANP on the LSTM (0.01); best-checkpoint test losses at $\varepsilon = 0.1/0.01/0.001$ were 29.18 / 2.23 / 0.36 (1.5-SPSA) and 2.31 / 1.57 / 0.20 (1.5-ANP) on MNIST, 2.41 / 2.37 / 1.60 and 2.41 / 2.30 / 1.78 on CIFAR-10, 3.66 / 2.09 / 2.05 and 2.47 / 2.14 / 2.27 on the LSTM, and 4.50 / 2.55 / 2.17 and 3.34 / 2.70 / 2.61 on the Transformer.

## F Selected configurations, throughput and ablation tables

Table 10 lists the selected $\varepsilon$ and the minute of the best checkpoint of every pair, Table 12 the best-validation-*accuracy* checkpoints of the two classification tasks, Table 11 the end-of-hour values,

Table 8: Tied $\varepsilon$ screening of every (task, method) pair: the lowest validation loss over the hour at each value (FF on the image tasks: goodness validation accuracy), with the selected value in bold. Values at $\varepsilon \leq 10^{-4}$ come from one-hour runs under the stage-2 settings; the first three columns are stage-1 runs where no stage-2 run exists. Every selected value has a worse neighbour on each side except where the grid ends at $10^{-5}$.

| Task | Method | $\varepsilon = 0.1$ | $\varepsilon = 0.01$ | $\varepsilon = 0.001$ | $\varepsilon = 0.0001$ | $\varepsilon = 10^{-5}$ |
|---|---|---|---|---|---|---|
| MNIST / MLP | BP | 1.826 | 2.744 | 0.362 | 0.138 | **0.094** |
| | SPSA | 2.566 | 2.087 | 0.293 | **0.121** | 0.185 |
| | 1.5-SPSA | 2.303 | 2.048 | 0.304 | **0.123** | 0.196 |
| | RS | 2.744 | 2.156 | 0.366 | **0.143** | 0.206 |
| | NP | 2.301 | 0.752 | 0.136 | **0.095** | 0.099 |
| | ANP | 2.304 | 0.446 | 0.120 | 0.100 | **0.100** |
| | 1.5-ANP | 2.304 | 1.454 | 0.180 | **0.132** | 0.140 |
| | INP | 2.305 | 0.396 | 0.128 | **0.093** | 0.101 |
| | DANP | 2.744 | 0.259 | **0.183** | 0.183 | – |
| | FF | 0.617 | **0.947** | 0.918 | 0.922 | – |
| CIFAR-10 / ConvNet | BP | 2.305 | 1.183 | 2.395 | **0.871** | 0.881 |
| | SPSA | 2.395 | 2.303 | 1.622 | **1.191** | 1.529 |
| | 1.5-SPSA | 2.395 | 2.303 | 1.604 | **1.213** | 1.563 |
| | RS | 2.395 | 2.327 | 1.738 | **1.276** | 1.591 |
| | NP | 2.395 | 2.395 | 1.400 | **1.059** | 1.222 |
| | ANP | 2.395 | 2.303 | 1.064 | **1.022** | 1.224 |
| | 1.5-ANP | 2.395 | 2.303 | 1.770 | 1.538 | **1.485** |
| | INP | 2.305 | 2.302 | 1.265 | **1.052** | 1.183 |
| | DANP | 2.395 | 1.129 | **1.052** | 1.158 | – |
| | FF | 0.224 | **0.338** | 0.334 | 0.135 | – |
| PTB / char-LSTM | BP | 1.765 | 1.315 | **1.268** | 1.492 | – |
| | SPSA | 3.597 | 2.117 | **2.066** | 2.993 | – |
| | 1.5-SPSA | 3.702 | 2.102 | **2.072** | 2.998 | – |
| | RS | 3.911 | 2.314 | **2.141** | 3.033 | – |
| | NP | 3.911 | 2.239 | **2.033** | 2.745 | – |
| | ANP | 3.911 | 3.382 | **2.970** | 2.999 | – |
| | 1.5-ANP | 2.494 | **2.189** | 2.287 | 2.518 | – |
| | INP | 2.652 | 2.007 | **1.971** | 2.563 | – |
| | DANP | 3.911 | 3.151 | **2.627** | 2.858 | – |
| | FF | 2.832 | **2.674** | 2.789 | 2.956 | – |
| PTB / char-Transformer | BP | 1.731 | 1.344 | **1.299** | 1.371 | – |
| | SPSA | 4.497 | 2.554 | **2.201** | 2.296 | – |
| | 1.5-SPSA | 4.497 | 2.561 | **2.172** | 2.301 | – |
| | RS | 4.497 | 2.728 | **2.297** | 2.347 | – |
| | NP | 4.497 | 2.957 | 2.359 | **2.353** | – |
| | ANP | 4.497 | 4.497 | 3.060 | **2.721** | – |
| | 1.5-ANP | 3.330 | 2.704 | 2.606 | **2.602** | – |
| | INP | 4.497 | 2.613 | **2.170** | 2.284 | – |
| | DANP | 4.497 | **2.530** | 2.530 | 2.996 | – |
| | FF | 3.046 | 2.924 | 2.946 | **2.919** | – |

Tables 13–15 the throughput, step counts and peak memory, and Tables 16–17 the per-layer alignment and the precision check.

## G Implementation details and hyper-parameters

**Perturbable layers.** Every trainable layer of every model is a perturbable linear, convolutional or embedding layer. A global context selects what a layer does on its forward pass: (i) a plain differentiable forward (backpropagation, forward-forward local losses, evaluation); (ii) a *clean* forward that detaches the layer input and records the input $\mathbf{h}_{l-1}$ and pre-activation $\mathbf{a}_l$, keeping a local

Table 9: Stage 1: final validation / test loss of all 96 runs of the original eight methods (seed 0, 5-minute evaluations). Bold: the $\varepsilon$ with the lowest *final* validation loss. A dagger marks a diverged run (last finite evaluation shown). Initial validation losses: 2.744 (MNIST), 2.395 (CIFAR-10), 3.911 (LSTM), 4.497 (Transformer).

| | | validation / test loss | | |
|---|---|---|---|---|
| Task | Optimizer | $\varepsilon = 0.1$ | $\varepsilon = 0.01$ | $\varepsilon = 0.001$ |
| MNIST / MLP | BP | **1.826 / 1.857** | 51.107 / 177.402$^\dagger$ | 2.239 / 2.355 |
| | SPSA | 957.457 / 2.302$^\dagger$ | 2.193 / 5.502 | **0.917 / 1.039** |
| | RS | 52.924 / 170.573$^\dagger$ | 8.157 / 4.833 | **1.238 / 1.412** |
| | NP | 2.301 / 2.301$^\dagger$ | 2.303 / 2.302 | **1.985 / 2.141** |
| | ANP | 2.323 / 2.319 | 0.519 / 0.619 | **0.363 / 0.381** |
| | INP | 2.325 / 2.321 | 8.917 / 21.010$^\dagger$ | **0.935 / 0.916** |
| | DANP | – / –$^\dagger$ | 1.033 / 1.190 | **0.643 / 0.751** |
| | FF | 2.288 / 2.288 | 2.243 / 2.241 | **2.123 / 2.127** |
| CIFAR-10 / ConvNet | BP | 2.305 / 2.305 | **1.357 / 2.402** | 23.735 / 24.475$^\dagger$ |
| | SPSA | 7.292 / 7.225$^\dagger$ | 22.978 / 2.313$^\dagger$ | **2.300 / 2.468** |
| | RS | **2.395 / 2.406** | 197.361 / 38.724$^\dagger$ | 2.401 / 2.492 |
| | NP | $1.9\times10^6$ / $1.3\times10^6$$^\dagger$ | 314.593 / 326.902$^\dagger$ | **1.597 / 1.613** |
| | ANP | 23.474 / 23.143$^\dagger$ | 2.332 / 2.336 | **1.143 / 1.177** |
| | INP | 2.327 / 2.323 | 2.303 / 2.303 | **1.321 / 1.339** |
| | DANP | – / –$^\dagger$ | **1.242 / 1.285** | 1.289 / 1.371 |
| | FF | 2.311 / 2.311 | 2.081 / 2.084 | **1.880 / 1.886** |
| PTB / char-LSTM | BP | 1.858 / 1.835 | 1.371 / 1.340 | **1.268 / 1.231** |
| | SPSA | 3.991 / 3.965 | 2.125 / 2.107 | **2.066 / 2.048** |
| | RS | 15.802 / 15.689$^\dagger$ | 2.330 / 2.310 | **2.141 / 2.124** |
| | NP | 23.036 / 22.815$^\dagger$ | 2.380 / 2.356 | **2.033 / 2.015** |
| | ANP | 195.280 / 192.326$^\dagger$ | 9.592 / 9.603$^\dagger$ | **3.242 / 3.232** |
| | INP | 2.885 / 2.866 | 2.007 / 1.989 | **1.971 / 1.955** |
| | DANP | 271.941 / 271.567$^\dagger$ | 13.384 / 13.251$^\dagger$ | **2.627 / 2.608** |
| | FF | 2.862 / 2.859 | **2.699 / 2.688** | 2.789 / 2.776 |
| PTB / char-Transformer | BP | 1.802 / 1.780 | 1.351 / 1.316 | **1.300 / 1.262** |
| | SPSA | 6.865 / 6.893 | 2.557 / 2.539 | **2.201 / 2.181** |
| | RS | 9.453 / 9.538$^\dagger$ | 2.745 / 2.726 | **2.297 / 2.279** |
| | NP | 6.617 / 6.746 | 3.206 / 3.180 | **2.374 / 2.357** |
| | ANP | 305.304 / 306.216$^\dagger$ | 27.783 / 27.601$^\dagger$ | **3.433 / 3.412** |
| | INP | 5.921 / 5.880 | 2.651 / 2.638 | **2.170 / 2.151** |
| | DANP | 26.685 / 26.551$^\dagger$ | **2.543 / 2.525** | 2.564 / 2.554 |
| | FF | 5.707 / 5.694 | **2.958 / 2.959** | 2.976 / 2.983 |

autograd graph to the layer's own parameters only; or (iii) a graph-free *noisy* forward that adds $\pm\sigma\boldsymbol{\xi}_l$ to the pre-activation of every layer (or of one selected layer for INP; the minus sign reuses the probes of the plus pass for the antithetic pairs of 1.5-ANP) and records $\boldsymbol{\xi}_l$ and $\tilde{\mathbf{a}}_l$. Given a per-unit credit tensor $\mathbf{G}_l$, the weight update $\mathbf{G}_l^\top \mathbf{h}_{l-1}$ and its convolutional and embedding analogues are obtained by a one-layer backward through the recorded local graph, so no error is propagated through any nonlinearity. A layer called several times in one forward (the LSTM gate layer, once per time step) accumulates one record per call.

**Decorrelation.** Each linear and convolutional layer of DANP owns a square matrix $\mathbf{R}_l$ (identity-initialised buffer, not an optimizer parameter) that multiplies the layer input; for convolutions the decorrelated vector is the unfolded $C_{\text{in}}k^2$ patch and the decorrelation is itself a convolution. $\mathbf{R}_l$ is updated once per step with Eq. (9) from the inputs recorded on the clean pass, from the first step. Embeddings are not decorrelated.

**1.5-ANP.** Attribution units are examples (images) or (sequence, position) tokens (text). The per-token loss is the next-character cross-entropy at that position; the per-token norm $\|\mathbf{d}_u\|^2$ sums the squared activity differences of every layer at that position (token embedding; the gate pre-activations

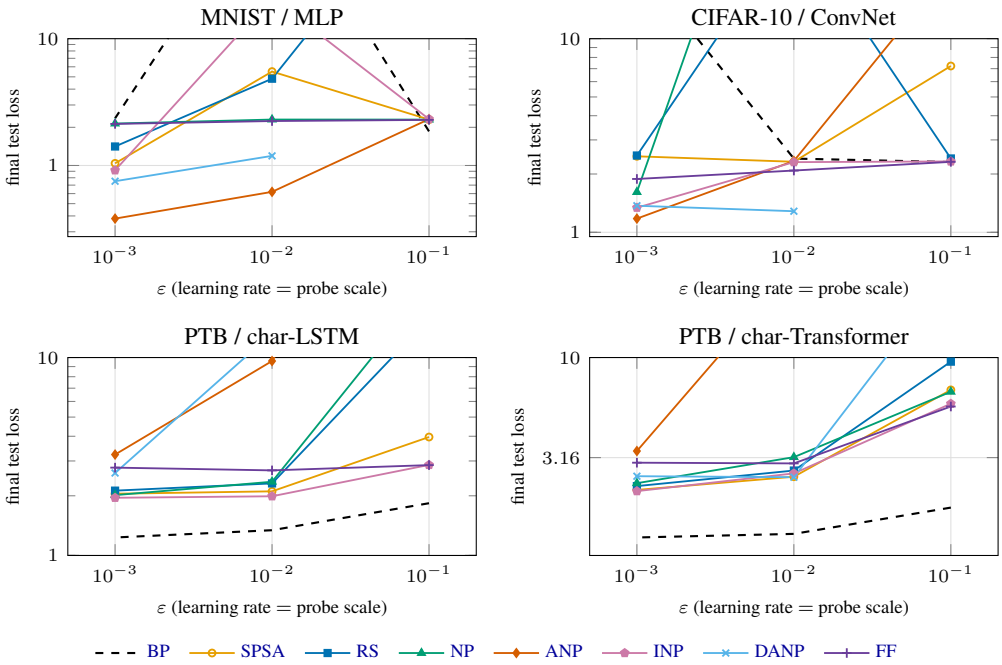



Figure 4: Stage 1: final test loss versus $\varepsilon$ (learning rate = probe scale) for the original eight methods; diverged runs are plotted at their last finite evaluation and the $y$-axis is capped at 10 (values up to $10^6$ leave the frame).

Table 10: Stage 2: selected $\varepsilon^*$, mean minute of the best validation checkpoint and number of seeds for every (task, method) pair.

| | MNIST | | CIFAR-10 | | LSTM | | Transformer | |
|---|---|---|---|---|---|---|---|---|
| Method | $\varepsilon^*$ | min / seeds | $\varepsilon^*$ | min / seeds | $\varepsilon^*$ | min / seeds | $\varepsilon^*$ | min / seeds |
| BP | 1e-05 | 3 / 1 | 0.0001 | 1 / 3 | 0.001 | 60 / 1 | 0.001 | 55 / 1 |
| SPSA | 0.0001 | 52 / 3 | 0.0001 | 60 / 3 | 0.001 | 60 / 1 | 0.001 | 60 / 1 |
| 1.5-SPSA | 0.0001 | 58 / 3 | 0.0001 | 59 / 3 | 0.001 | 60 / 3 | 0.001 | 59 / 3 |
| RS | 0.0001 | 46 / 3 | 0.0001 | 59 / 3 | 0.001 | 60 / 1 | 0.001 | 60 / 1 |
| NP | 0.0001 | 6 / 3 | 0.0001 | 59 / 1 | 0.001 | 60 / 1 | 0.0001 | 60 / 1 |
| ANP | 1e-05 | 48 / 1 | 0.0001 | 59 / 1 | 0.001 | 25 / 1 | 0.0001 | 19 / 1 |
| 1.5-ANP | 0.0001 | 29 / 3 | 1e-05 | 60 / 1 | 0.01 | 10 / 3 | 0.0001 | 27 / 1 |
| INP | 0.0001 | 7 / 3 | 0.0001 | 54 / 1 | 0.001 | 60 / 1 | 0.001 | 60 / 1 |
| DANP | 0.001 | 1 / 3 | 0.001 | 8 / 1 | 0.001 | 60 / 1 | 0.01 | 50 / 1 |
| FF | 0.01 | 42 / 3 | 0.01 | 18 / 1 | 0.01 | 20 / 1 | 0.0001 | 56 / 1 |

of the LSTM call at that time step; the projections and feed-forward layers of the Transformer at that position; the readout), and the position embedding, perturbed once per batch, contributes its row for that position to every sequence and receives the sum over sequences of $\beta_u$. The noisy passes run the perturbed recurrent state without the specification's 0.5 damping. $\lambda = 10^{-3}$, $\alpha = 0.5$, $\epsilon_{\text{mach}} = 10^{-12}$.

**Language-model specifics.** The LSTM is written with a single perturbable layer for the four gate pre-activations so that the gates are the perturbable nodes. The Transformer uses pre-LN blocks with parameter-free layer normalisation, a learned positional embedding over $T = 64$ positions, and a final parameter-free layer normalisation before the readout. The $n$-gram reference points of Section 5.7 are count models estimated on the training characters and evaluated on the test characters: uniform over $V = 50$, maximum-likelihood unigram and bigram, and a trigram with add-0.1 smoothing.

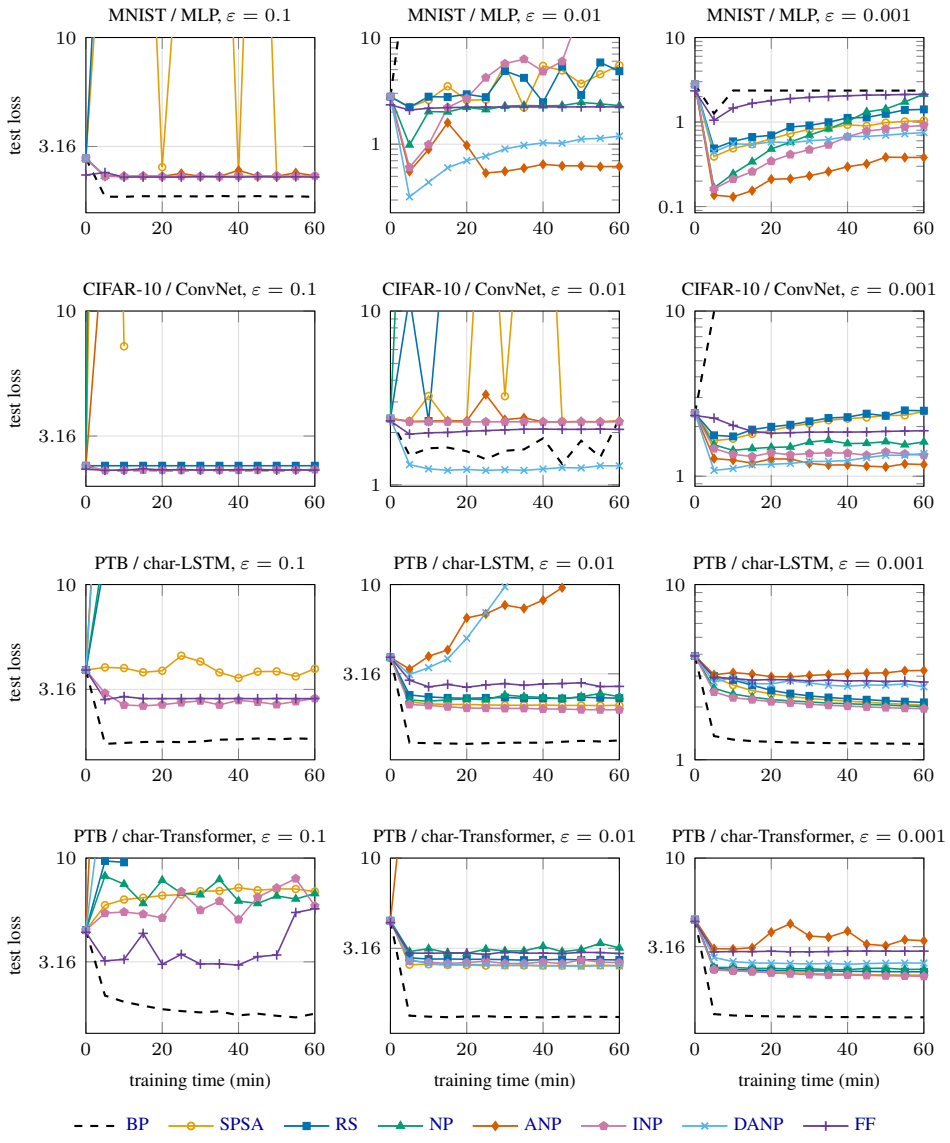



Figure 5: Stage 1: test loss versus training time for every run of the original eight methods. Rows: tasks; columns: $\varepsilon \in \{0.1, 0.01, 0.001\}$.

**Forward-forward specifics.** One Adam instance updates all parameters; every block's goodness loss is computed on the detached, normalised output of the previous block, so each parameter receives exactly one local gradient per step (the embeddings are trained by the first block's goodness). The label overlay follows Pezeshki [2023]: the first 10 entries of the flattened input are zeroed and the entry of the label is set to the maximum input value (for CIFAR-10, the first 10 pixels of the first row of channel 0); a wrong label is drawn uniformly from the nine other classes; the neutral label sets all 10 entries to 0.1 times the maximum. For PTB, 30 % of the positions of each sequence are replaced by uniformly random characters, resampled every step. Label accumulation sums the goodness over all blocks but the first (Hinton) or over all blocks (logged as a second number). The readout is trained every step by cross-entropy on the normalised, detached top-block activity.

**Evaluation and checkpoints.** Validation and test metrics are computed on the full splits at initialisation, every minute of training time (every 5 minutes in stage 1) and at the end; the clock is paused during evaluation. Stage 2 saves the parameters at the best validation loss, at 5 minutes and at the end

Table 11: The same runs at the end of the hour (secondary, stability metric): test loss and accuracy or bits per character after 60 minutes of training. A dagger marks a pair in which at least one seed was stopped by the divergence rule (its last evaluation is used).

| | MNIST / MLP | | CIFAR-10 / ConvNet | | PTB / char-LSTM | | PTB / char-Transformer | |
|---|---|---|---|---|---|---|---|---|
| Method | loss | acc. (%) | loss | acc. (%) | loss | bits/char | loss | bits/char |
| BP | 1.057 | 96.6 | 14.311(2756) | 65.6(3) | 1.231$^{s1}$ | 1.78 | 1.262$^{s1}$ | 1.82 |
| SPSA | 0.139(5) | 96.0(2) | 1.200(50) | 57.7(18) | 2.048$^{s1}$ | 2.96 | 2.181$^{s1}$ | 3.15 |
| 1.5-SPSA | 0.141(6) | 95.9(2) | 1.216(47) | 56.6(18) | 2.054(8) | 2.96(1) | 2.154(25) | 3.11(4) |
| RS | 0.161(9) | 95.2(2) | 1.295(55) | 53.5(23) | 2.124$^{s1}$ | 3.06 | 2.279$^{s1}$ | 3.29 |
| NP | 0.344(70) | 97.0(0) | 1.096 | 61.5 | 2.015$^{s1}$ | 2.91 | 2.336 | 3.37 |
| ANP | **0.107** | **96.8** | **1.063** | **63.4** | 3.232$^{s1}$ | 4.66 | 2.923 | 4.22 |
| 1.5-ANP | 0.163(5) | 96.2(4) | 1.491 | 47.3 | 2.368(68) | 3.42(10) | 2.787 | 4.02 |
| INP | 0.317(65) | 96.9(2) | 1.079 | 62.4 | **1.955**$^{s1}$ | **2.82** | **2.151**$^{s1}$ | **3.10** |
| DANP | 0.782(24) | 94.2(2) | 1.378 | 64.8 | 2.608$^{s1}$ | 3.76 | 2.525$^{s1}$ | 3.64 |
| FF | 2.300(92) | 93.1(3) [14.2] | 2.103 | 32.1 [23.1] | 2.688$^{s1}$ | 3.88 | 2.923 | 4.22 |

Table 12: Image tasks: test accuracy (%) and test loss at the evaluation with the highest validation *accuracy*, mean over seeds. Best probe method per task in bold.

| | MNIST / MLP | | CIFAR-10 / ConvNet | |
|---|---|---|---|---|
| Method | acc. (%) | loss | acc. (%) | loss |
| BP | 97.1 | 0.102 | 70.0(4) | 0.964(53) |
| SPSA | 95.8(2) | 0.143(9) | 57.7(18) | 1.203(50) |
| 1.5-SPSA | 95.7(2) | 0.145(9) | 56.4(16) | 1.222(44) |
| RS | 95.0(3) | 0.164(10) | 53.6(21) | 1.298(52) |
| NP | **97.1(2)** | 0.128(29) | 62.1 | 1.081 |
| ANP | 96.7 | 0.109 | 63.1 | 1.060 |
| 1.5-ANP | 96.0(4) | 0.161(7) | 46.7 | 1.507 |
| INP | 97.0(1) | 0.242(61) | 62.1 | 1.096 |
| DANP | 94.0(0) | 0.490(246) | **63.2** | 1.266 |
| FF | 93.1(3) | 2.252(34) | 33.6 | 2.032 |

(released with the logs). Runs are not bitwise reproducible from the seed because cuDNN autotuning is on and stopping is time-based.

**Hyper-parameters.** Table 18 summarises every hyper-parameter; none is tuned per method beyond the $\varepsilon$ screening.

**Data and licences.** MNIST [LeCun et al., 1998], CIFAR-10 [Krizhevsky, 2009] and the character-level Penn Treebank files [Marcus et al., 1993, Mikolov, 2012] are used as distributed by their authors. MNIST is distributed under the Creative Commons Attribution-Share Alike 3.0 licence; CIFAR-10 is released by its authors for research use without a formal licence; the character-level PTB files are the widely redistributed Mikolov preprocessing of the Penn Treebank (LDC99T42) and are used for research only. Our code is released under the MIT licence.

**Compute.** Stage 1 used about 100 GPU-hours (10.5 hours of wall-clock on the 8-GPU node); stage 2 and the ablations about 500 GPU-hours (24 hours of wall-clock on up to five 8-GPU nodes); the alignment diagnostic takes about an hour on one GPU. Stage 1 was preceded by 32 two-minute smoke runs and four 10-minute backpropagation runs with 30-second evaluations (released with the logs, superseded by Table 6).

Table 13: Throughput: optimizer steps and forward passes per hour (thousands), mean over the seeds of the selected runs, measured with eight runs sharing one host (one BP step is counted as one pass; its backward is not counted).

| | MNIST | | CIFAR-10 | | LSTM | | Transformer | |
|---|---|---|---|---|---|---|---|---|
| Method | steps/h | passes/h | steps/h | passes/h | steps/h | passes/h | steps/h | passes/h |
| BP | 4278.4 | 4278.4 | 2023.8 | 2023.8 | 147.5 | 147.5 | 1021.0 | 1021.0 |
| SPSA | 1002.7 | 16043.3 | 590.4 | 9446.4 | 34.3 | 548.1 | 219.1 | 3506.3 |
| 1.5-SPSA | 850.3 | 14455.1 | 511.2 | 8690.2 | 33.6 | 571.8 | 227.2 | 3862.4 |
| RS | 1075.8 | 17212.5 | 609.0 | 9743.7 | 34.5 | 552.7 | 229.1 | 3665.6 |
| NP | 587.4 | 9397.7 | 235.9 | 3773.9 | 23.6 | 377.4 | 74.7 | 1195.1 |
| ANP | 493.3 | 7892.1 | 177.6 | 2842.0 | 19.8 | 317.2 | 62.5 | 999.6 |
| 1.5-ANP | 502.9 | 8549.5 | 351.9 | 5983.1 | 23.5 | 399.0 | 73.6 | 1251.2 |
| INP | 598.7 | 9579.0 | 283.6 | 4537.6 | 24.9 | 398.9 | 168.0 | 2688.0 |
| DANP | 430.3 | 6885.3 | 243.3 | 3892.3 | 17.0 | 272.5 | 86.8 | 1388.1 |
| FF | 1685.1 | 5055.3 | 941.0 | 2823.0 | 139.7 | 279.4 | 271.0 | 542.0 |

Table 14: Steps completed in the hour (thousands) and milliseconds of training time per step, mean over seeds of the selected runs (eight runs sharing one host).

| | MNIST | | CIFAR-10 | | LSTM | | Transformer | |
|---|---|---|---|---|---|---|---|---|
| Method | steps (k) | ms/step | steps (k) | ms/step | steps (k) | ms/step | steps (k) | ms/step |
| BP | 4278 | 0.84 | 2024(761) | 2.02 | 148 | 24.41 | 1021 | 3.53 |
| SPSA | 1003(312) | 3.89 | 590(173) | 6.55 | 34 | 105.08 | 219 | 16.43 |
| 1.5-SPSA | 850(264) | 4.59 | 511(149) | 7.56 | 34(1) | 107.12 | 227(2) | 15.85 |
| RS | 1076(357) | 3.68 | 609(177) | 6.34 | 35 | 104.22 | 229 | 15.71 |
| NP | 587(163) | 6.53 | 236 | 15.26 | 24 | 152.61 | 75 | 48.20 |
| ANP | 493 | 7.30 | 178 | 20.27 | 20 | 181.58 | 62 | 57.62 |
| 1.5-ANP | 503(161) | 7.81 | 352 | 10.23 | 23(0) | 153.44 | 74 | 48.91 |
| INP | 599(188) | 6.54 | 284 | 12.69 | 25 | 144.40 | 168 | 21.43 |
| DANP | 430(7) | 8.37 | 243 | 14.80 | 17 | 211.40 | 87 | 41.49 |
| FF | 1685(17) | 2.14 | 941 | 3.83 | 140 | 25.77 | 271 | 13.28 |

Table 15: Peak GPU memory of the training process (MB): tensors allocated, and reserved by the caching allocator, recorded during the stage-2 runs.

| | MNIST | | CIFAR-10 | | LSTM | | Transformer | |
|---|---|---|---|---|---|---|---|---|
| Method | alloc. | reserved | alloc. | reserved | alloc. | reserved | alloc. | reserved |
| BP | 243 | 258 | 1889 | 1972 | – | – | – | – |
| SPSA | 235 | 258 | 1889 | 1972 | – | – | – | – |
| 1.5-SPSA | 235 | 258 | 1889 | 1972 | 118 | 142 | 321 | 420 |
| RS | 235 | 258 | 1889 | 1972 | – | – | – | – |
| NP | 243 | 258 | 1889 | 1972 | – | – | 369 | 482 |
| ANP | 243 | 258 | 1889 | 1972 | – | – | 369 | 490 |
| 1.5-ANP | 243 | 258 | 1889 | 1972 | 148 | 178 | 369 | 538 |
| INP | 243 | 258 | 1889 | 1972 | – | – | – | – |
| DANP | 247 | 278 | – | – | – | – | – | – |
| FF | 249 | 278 | – | – | – | – | 353 | 430 |

Table 16: Per-layer cosine between each estimator and the true gradient at 1024 forward passes, $\sigma = 10^{-3}$, after the 2000-step backpropagation warm-up (layer indices in module order; "all" is the whole parameter vector).

| Model | Layer | SPSA | RS | NP | ANP | INP | DANP | 1.5-SPSA | 1.5-ANP |
|---|---|---|---|---|---|---|---|---|---|
| MNIST / MLP | 0 | 0.08 | 0.05 | 0.95 | 0.94 | 0.93 | 0.94 | 0.08 | 0.48 |
| | 1 | 0.30 | 0.23 | 0.95 | 0.90 | 0.96 | 0.89 | 0.30 | 0.78 |
| | 2 | 0.56 | 0.35 | 0.98 | 0.89 | 0.99 | 0.93 | 0.56 | 0.83 |
| | all | 0.11 | 0.07 | 0.96 | 0.81 | 0.95 | 0.83 | 0.11 | 0.69 |
| CIFAR-10 / ConvNet | 0 | 0.65 | 0.41 | 0.56 | 0.55 | 0.46 | 0.44 | 0.65 | -0.10 |
| | 1 | 0.15 | 0.10 | 0.43 | 0.54 | 0.48 | 0.54 | 0.15 | 0.10 |
| | 2 | 0.10 | 0.05 | 0.45 | 0.62 | 0.60 | 0.60 | 0.10 | 0.24 |
| | 3 | 0.04 | 0.04 | 0.81 | 0.93 | 0.93 | 0.86 | 0.04 | 0.87 |
| | 4 | 0.21 | 0.15 | 0.94 | 0.88 | 0.99 | 0.85 | 0.21 | 0.87 |
| | all | 0.12 | 0.08 | 0.53 | 0.52 | 0.51 | 0.53 | 0.12 | 0.27 |
| PTB / char-LSTM | 0 | 0.12 | 0.11 | 0.58 | 0.58 | 0.41 | 0.97 | 0.12 | 0.42 |
| | 1 | 0.08 | 0.05 | 0.07 | 0.04 | 0.16 | 0.17 | 0.08 | 0.21 |
| | 2 | 0.23 | 0.15 | 0.13 | 0.05 | 0.27 | 0.49 | 0.23 | 0.36 |
| | all | 0.11 | 0.07 | 0.08 | 0.04 | 0.18 | 0.22 | 0.11 | 0.26 |
| PTB / char-Transformer | 0 | 0.21 | 0.09 | 0.44 | 0.45 | 0.22 | 0.81 | 0.21 | 0.45 |
| | 1 | 0.15 | 0.10 | 0.34 | 0.33 | 0.12 | 0.42 | 0.15 | 0.20 |
| | 2 | 0.06 | -0.02 | 0.07 | 0.15 | 0.12 | 0.10 | 0.06 | 0.18 |
| | 3 | 0.13 | 0.10 | 0.04 | 0.50 | 0.11 | 0.35 | 0.13 | 0.38 |
| | 4 | 0.11 | 0.06 | 0.03 | 0.20 | 0.23 | 0.45 | 0.11 | 0.26 |
| | 5 | 0.17 | 0.12 | 0.13 | 0.16 | 0.18 | 0.40 | 0.17 | 0.15 |
| | 6 | 0.10 | 0.11 | 0.00 | 0.12 | 0.21 | 0.39 | 0.11 | 0.14 |
| | 7 | 0.10 | 0.07 | -0.01 | 0.23 | 0.19 | 0.54 | 0.10 | 0.23 |
| | 8 | 0.03 | 0.01 | 0.05 | 0.20 | 0.32 | 0.06 | 0.03 | 0.15 |
| | 9 | 0.08 | 0.02 | 0.07 | 0.03 | 0.16 | 0.07 | 0.08 | -0.03 |
| | 10 | 0.11 | 0.08 | 0.10 | 0.13 | 0.15 | 0.34 | 0.11 | 0.09 |
| | 11 | 0.07 | 0.08 | 0.07 | 0.11 | 0.23 | 0.46 | 0.07 | 0.26 |
| | 12 | 0.04 | 0.02 | -0.02 | 0.09 | 0.10 | 0.27 | 0.05 | 0.21 |
| | 13 | 0.07 | 0.01 | 0.07 | 0.20 | 0.12 | 0.53 | 0.08 | 0.33 |
| | 14 | 0.04 | 0.01 | -0.02 | 0.16 | 0.13 | 0.40 | 0.04 | 0.24 |
| | all | 0.11 | 0.07 | 0.05 | 0.07 | 0.17 | 0.13 | 0.11 | 0.09 |

Table 17: Precision check on the ConvNet: whole-model cosine with the true gradient at 1024 passes in fp32 with TF32 off, fp32 with TF32 on (PyTorch's default for convolutions) and fp64, same probes.

| Model | Estimator | $\sigma$ | fp32 | TF32 | fp64 |
|---|---|---|---|---|---|
| CIFAR-10 / ConvNet | NP | 0.01 | 0.442 | 0.440 | 0.459 |
| | NP | 0.001 | 0.445 | 0.420 | 0.475 |
| | ANP | 0.01 | 0.189 | 0.189 | 0.193 |
| | ANP | 0.001 | 0.481 | 0.479 | 0.482 |
| | INP | 0.01 | 0.442 | 0.439 | 0.477 |
| | INP | 0.001 | 0.425 | 0.358 | 0.484 |
| | SPSA | 0.01 | 0.115 | 0.115 | 0.108 |
| | SPSA | 0.001 | 0.116 | 0.116 | 0.110 |

Table 18: Hyper-parameters shared by all runs.

| | |
|---|---|
| Optimizer | Adam, $(\beta_1, \beta_2) = (0.9, 0.99)$, $\epsilon_{\text{Adam}} = 10^{-8}$, no schedule, no weight decay (AdamW $10^{-2}$ in the BP ablation only) |
| Learning rate $\eta$ = probe scale $\sigma = \varepsilon$ | $\{0.1, 0.01, 0.001, 0.0001\}$, $10^{-5}$ on the image tasks (Table 8) (untied grid in Section 5.4; own recipes in Section 5.5) |
| Probe distribution | $\mathcal{N}(\mathbf{0}, \mathbf{I})$, scaled by $\sigma$ |
| Forward passes per step | 16 (probe methods), 17 (1.5-SPSA, 1.5-ANP), 3 / 2 (FF), 1 + 1 backward (BP) |
| SPSA, 1.5-SPSA pairs / RS candidates | 8 / 15 |
| Curvature exponent $\alpha$ / floor $\lambda$ | 1.5-SPSA: 0.1 / 1.0; 1.5-ANP: 0.5 / $10^{-3}$ |
| Batch size | 128 images; 64 sequences $\times$ 64 characters |
| Initialisation | Xavier-uniform, zero biases (LSTM forget gate = 1) |
| FF threshold / PTB corruption | 2 (0.25 for the LSTM block) / 30 % of characters |
| Decorrelation rate $\eta_R$ (DANP) | $10^{-3}$, fixed |
| Training budget | 60 min of training time per run, evaluation excluded (30 min in the decoupling ablation) |
| Evaluation interval | every 1 min (stage 2) / 5 min (stage 1) of training time, and at the end |
| Divergence rule | non-finite loss, or validation loss $> 10\times$ its step-0 value at two consecutive evaluations (stage 1: $2\times$, plus the training-loss clause) |
| Precision | fp32, TF32 off (stage 2); PyTorch default, TF32 on for convolutions (stage 1) |
| Seeds | 3 (stage 2 and the replicated untied / own-recipe settings), 1 (stage 1 and screenings) |

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